Subcubic homomorphism counts determine every graph of nullity at most one

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Editorial note. Complete original mathematical paper and all three original proof sections, native TikZ figures and references.bib retained. source0 expands all TeX input passages verbatim; original files and original-to-render locator map preserved. No mathematical changes, added assumptions, author review or independent certification.

Editorial note. Original declaration2582–2590 retained: author formulated problem and initial statements; GPT-6 Astra assisted exploration/literature and proofs developed through discussions with author; Astra/Claude Opus5.5 helped readability/presentation, author assumes responsibility. No explicit selected-core discovery attribution is inferred and no AI-free assertion is made.

Editorial note. Only the three principal independent objects are counted; all supporting tensor characterization, attachments and interpolation gadgets remain complete proof support.

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Allowed standard prerequisites (editorial).

- Graph homomorphism counts, adjacency spectra, CFI graphs and tensor/network interpolation. Native finite simple undirected graph conventions, homomorphism indistinguishability, nullity and all exact local constructions retained. Prior nonsingular results and CFI/minor-closed indistinguishability inputs retain their original labels and bibliographic pointers.. Full paper and source-local auxiliary proofs/TikZ figures accompany each selected principal theorem; no independent proof reconstruction or certification.

1 Introduction and Main Results

Graphs are finite, simple and undirected unless stated otherwise. We write $N_G(x)$ for the set of neighbours of x in G and $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ for the set of nonzero real numbers. The *nullity* of a graph is the dimension of the kernel of its adjacency matrix. A *homomorphism* from a graph F to a graph G is a map from $V(F)$ to $V(G)$ that preserves adjacency. We write $\hom(F, G)$ for the number of such maps. Two graphs G and G' are *homomorphism indistinguishable* over a graph class $\mathcal{C}$, written $G \equiv_{\mathcal{C}} G'$, if $\hom(F, G) = \hom(F, G')$ for every $F \in \mathcal{C}$.

Homomorphism counts characterise a range of graph properties and equivalence relations. Lovász [12] proved that two graphs G and G' are isomorphic if and only if $\hom(F, G) = \hom(F, G')$ for every graph F. A graph is bipartite if and only if its homomorphism count from every odd cycle is zero. Two graphs G and G' with the same number of vertices are cospectral if and only if $\hom(C, G) = \hom(C, G')$ for every cycle C, because the homomorphism count from the k-cycle to a graph is the trace of the k-th power of its adjacency matrix. For every $k \geq 1$, two graphs are homomorphism indistinguishable over the graphs of treewidth at most k if and only if the k-dimensional Weisfeiler–Leman test does not distinguish them [7, 5].

A graph has *simple spectrum* if its adjacency matrix has pairwise distinct eigenvalues. We study homomorphism characterisations of isomorphism on graphs with simple spectrum. We call a class *complete on simple spectrum* if any two graphs with simple spectrum that are homomorphism indistinguishable over the class are isomorphic. By the result of Lovász, the class of all graphs is complete on simple spectrum. Since isomorphism testing on graphs with simple spectrum is possible in polynomial time [2], we ask which smaller classes are still complete on simple spectrum.

We first prove that the following graph classes are not complete on simple spectrum.

Theorem 1. *Let $\mathcal{C}$ be any one of the following graph classes:*

1. *graphs of treewidth at most k, for a fixed integer $k \geq 1$;*
2. *graphs embeddable on a surface of Euler genus at most g, for a fixed integer $g \geq 0$;*
3. *graphs with no K_7 minor.*

There are non-isomorphic graphs G, G' with simple spectrum such that $G \equiv_{\mathcal{C}} G'$. The graphs G and G' can be chosen cospectral with each other, with each of them bipartite, nonsingular, and of maximum degree at most six. Consequently every class complete on simple spectrum has unbounded treewidth and unbounded Euler genus, and contains a graph with a K_7 minor.

The first case of Theorem 1 answers the question of Hordan, Dym and Seppelt [11, Section 7], who proved this incompleteness for multigraphs and asked whether it holds for simple graphs. The second case with $g = 0$ covers planar graphs. Planar graphs have unbounded treewidth, so unbounded treewidth alone does not suffice for completeness. The proof attaches rooted trees to a pair of Cai–Fürer–Immerman graphs. The two resulting graphs have simple spectrum and are homomorphism indistinguishable over every minor-closed class over which the original pair is homomorphism indistinguishable (Lemma 4). The two Cai–Fürer–Immerman graphs are homomorphism indistinguishable from each other over each of the three classes by known results.

It remains open whether every class complete on simple spectrum must contain graphs with arbitrarily large clique minors. An affirmative answer would imply the weak Roberson conjecture, which states that for every proper minor-closed graph class there are two non-isomorphic graphs that are homomorphism indistinguishable over the class [15, Conjecture 5 and Theorem 8.4]; see Remark 11.

We next ask which graph classes suffice for completeness on simple spectrum. A graph is *subcubic* if its maximum degree is at most three. Let $\mathcal{D}_3$ be the class of subcubic graphs. There are non-isomorphic graphs that are homomorphism indistinguishable over $\mathcal{D}_3$ [15]. Our next result shows that subcubic homomorphism counts nevertheless determine every graph of nullity at most one up to isomorphism. A graph with simple spectrum has nullity at most one, so the result covers every graph with simple spectrum.

Theorem 2. *Let G be a graph of nullity at most one. For every graph G', $G \equiv_{\mathcal{D}_3} G'$ if and only if $G \cong G'$. In particular, $\mathcal{D}_3$ is complete on simple spectrum.*

Theorem 2 extends [4, Corollary 38] from nonsingular graphs to graphs of nullity at most one. Lemma 13 characterises subcubic homomorphism indistinguishability for arbitrary graphs G, G': $G \equiv_{\mathcal{D}_3} G'$ if and only if G and G' have the same number of vertices and there is a bijective linear isometry between the images of their adjacency matrices that intertwines the adjacency operators and preserves the coordinatewise trilinear forms. For degree bound three, Lemma 13 is the more specific characterisation suggested after Corollary 39 of [4].

It remains open whether Theorem 2 extends to nullity two, but there are counterexamples of nullity three; see Remark 17.

Finally, we show that homomorphism counts from subdivisions of suitable cubic graphs, together with the number of vertices, have the same distinguishing power as counts from all subcubic graphs.

Proposition 3. *Let $(H_N)_{N \geq 1}$ be a sequence of connected cubic graphs such that, for every $t \geq 1$, some H_N contains K_t as a minor. Let $\mathcal{F}$ consist of K_1 and all subdivisions of the graphs H_N. Then*

$$G \equiv_{\mathcal{F}} G' \iff G \equiv_{\mathcal{D}_3} G' \tag{1}$$

for all graphs G, G'. In particular, $\mathcal{F}$ is complete on simple spectrum.

Including K_1 in $\mathcal{F}$ is necessary: K_2 and $K_2 \sqcup K_1$ have simple spectrum and equal homomorphism counts from every connected graph with at least two vertices.

2 Proof of Theorem 1

We use the variant of the Cai–Fürer–Immerman (CFI) construction [3] given in [15, Definition 3.1]. Let B be a connected graph and v_* a vertex of B. For a vertex v of B, the set of edges incident with v is denoted by $E_B(v)$. For $i \in \{0,1\}$, let $\mathrm{CFI}_i(B)$ be the graph with

$$\begin{aligned} \Xi_i(v) &= \left\{ \xi \in \mathbb{F}_2^{E_B(v)} : \sum_{e \in E_B(v)} \xi_e = i\delta_{v,v_*} \right\}, \\ V(\mathrm{CFI}_i(B)) &= \{(v,\xi) : v \in V(B), \xi \in \Xi_i(v)\}, \\ E(\mathrm{CFI}_i(B)) &= \{(v,\xi)(w,\zeta) : vw \in E(B), \xi \in \Xi_i(v), \zeta \in \Xi_i(w), \xi_{vw} = \zeta_{vw}\}. \end{aligned} \tag{2}$$

Here δ_{ab} is 1 when $a = b$ and 0 otherwise. The vertices (v,ξ) with $\xi \in \Xi_i(v)$ form a *fibre* with *base vertex* v, and they are said to lie *over* v. The isomorphism type of $\mathrm{CFI}_1(B)$ is independent of the choice of v_*, and the graphs $\mathrm{CFI}_0(B)$ and $\mathrm{CFI}_1(B)$ are non-isomorphic; both facts follow from [13, Lemma 5].

For $i \in \{0,1\}$, let $\pi_i : \mathrm{CFI}_i(B) \to B$ be the projection $\pi_i(v,\xi) = v$. Let F be a graph and $\chi : F \to B$ a homomorphism. A homomorphism $f : F \to \mathrm{CFI}_i(B)$ is a *lift* of χ if $\pi_i \circ f = \chi$. The number of such lifts is denoted by $\hom_\chi(F, \mathrm{CFI}_i(B))$. We show that if $\hom(F, \mathrm{CFI}_0(B)) = \hom(F, \mathrm{CFI}_1(B))$, then $\hom_\chi(F, \mathrm{CFI}_0(B)) = \hom_\chi(F, \mathrm{CFI}_1(B))$ for every homomorphism $\chi : F \to B$. For every fixed χ, [15, Theorem 3.6] gives

$$\hom_\chi(F, \mathrm{CFI}_1(B)) \le \hom_\chi(F, \mathrm{CFI}_0(B)). \tag{3}$$

Every homomorphism $f : F \to \mathrm{CFI}_i(B)$ is a lift of exactly one map $\chi = \pi_i \circ f$, so

$$\hom(F, \mathrm{CFI}_0(B)) - \hom(F, \mathrm{CFI}_1(B)) = \sum_\chi \big(\hom_\chi(F, \mathrm{CFI}_0(B)) - \hom_\chi(F, \mathrm{CFI}_1(B))\big) \tag{4}$$

is a sum of nonnegative terms. If the total counts agree, every term therefore vanishes. Consequently, for every graph F and every homomorphism $\chi : F \to B$,

$$\hom(F, \mathrm{CFI}_0(B)) = \hom(F, \mathrm{CFI}_1(B)) \Longrightarrow \hom_\chi(F, \mathrm{CFI}_0(B)) = \hom_\chi(F, \mathrm{CFI}_1(B)). \tag{5}$$

For a minor-closed class $\mathcal{C}$, the following lemma constructs non-isomorphic graphs $G \equiv_{\mathcal{C}} G'$ with simple spectrum whenever $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$.

Lemma 4. *Let $\mathcal{C}$ be a minor-closed graph class and let B be a connected graph satisfying $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$. Then there are non-isomorphic graphs G, G' with simple spectrum and $G \equiv_{\mathcal{C}} G'$. The witnesses can be chosen bipartite, cospectral, nonsingular, and of maximum degree at most six.*

Proof of Theorem 1. By Lemma 4, it suffices to give, for each of the three classes $\mathcal{C}$, a connected graph B with $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$. When $\mathcal{C}$ is the class of graphs of treewidth at most $k \ge 1$, the graph $B = K_{k+2}$ satisfies $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$ [13, Theorem 10 and Lemma 12]. When $\mathcal{C}$ is the class of graphs of Euler genus at most g, a connected graph B with $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$ is given in [14, Lemma 2.10 and proof of Theorem 1.8]. When $\mathcal{C}$ is the class of K_7-minor-free graphs, a connected graph B with $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$ is given in [14, Lemma 2.10 and proof of Theorem 1.9] and [16, Theorem 1.7 and Section 1.3]. □

The proof of Lemma 4 has two steps. If $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$ for a minor-closed class $\mathcal{C}$, then the two graphs obtained from $\mathrm{CFI}_0(B)$ and $\mathrm{CFI}_1(B)$ by attaching the same rooted graph to every vertex of a fibre are again homomorphism indistinguishable over $\mathcal{C}$ (Lemma 5). For a suitable choice of attached trees, the resulting graphs have simple spectrum (Lemma 10). The second step uses two lemmas on the attached trees (Lemmas 6 and 8).

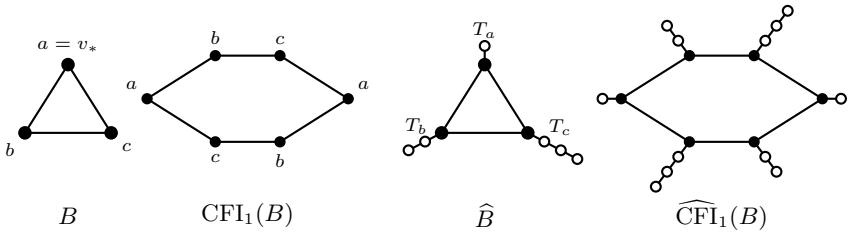


Figure 1: A CFI-compatible rooted product for $B = K_3$ with $v_* = a$. The hexagon labels indicate the base vertices of the fibres. The attached paths T_a, T_b, T_c have lengths one, two and three; each vertex in the fibre over z receives a copy of T_z.

Attaching a copy of a rooted graph at a vertex means identifying the root of the copy with that vertex. Fix a connected base graph B. For each $z \in V(B)$, let T_z be a rooted graph with root r_z. Let the *rooted product* $\widehat{B}$ be the graph obtained by attaching one copy of T_z at each $z \in V(B)$ and retaining all edges of B. For $i \in \{0,1\}$, let the *CFI-compatible rooted product* $\widehat{\mathrm{CFI}}_i(B)$ be the graph obtained by attaching a separate copy of T_v at each $(v,\xi) \in V(\mathrm{CFI}_i(B))$ and retaining all edges of $\mathrm{CFI}_i(B)$. In both products, each attached copy includes its root. The projection π_i extends to $\widehat{\pi}_i : \widehat{\mathrm{CFI}}_i(B) \to \widehat{B}$ by mapping each attached copy of T_z identically onto the copy of T_z in $\widehat{B}$. Figure 1 shows B, $\mathrm{CFI}_1(B)$ and their rooted products for a base graph with three vertices.

Lemma 5. *Let $\mathcal{C}$ be a minor-closed graph class and B a connected graph such that $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$. For any family of rooted graphs $(T_z)_{z\in V(B)}$, the CFI-compatible rooted products satisfy $\widehat{\mathrm{CFI}}_0(B) \equiv_{\mathcal{C}} \widehat{\mathrm{CFI}}_1(B)$.*

Proof. Fix $F \in \mathcal{C}$. We prove that, for $i \in \{0,1\}$,

$$\hom(F, \widehat{\mathrm{CFI}}_i(B)) = \sum_{\eta : F \to \widehat{B}} \hom_{\chi_\eta}(Q_\eta, \mathrm{CFI}_i(B)), \tag{6}$$

where, for each homomorphism $\eta : F \to \widehat{B}$, the graph Q_η and the homomorphism $\chi_\eta : Q_\eta \to B$ are defined below. The graph Q_η is obtained from F by contracting the connected components of a subgraph of F and replacing parallel edges by single edges, so Q_η is a minor of F, hence $Q_\eta \in \mathcal{C}$ because $\mathcal{C}$ is minor-closed. Since $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$ and $Q_\eta \in \mathcal{C}$, we have $\hom(Q_\eta, \mathrm{CFI}_0(B)) = \hom(Q_\eta, \mathrm{CFI}_1(B))$. By Equation (5), $\hom_{\chi_\eta}(Q_\eta, \mathrm{CFI}_0(B)) = \hom_{\chi_\eta}(Q_\eta, \mathrm{CFI}_1(B))$. Assuming Equation (6), summing these equalities over η gives $\hom(F, \widehat{\mathrm{CFI}}_0(B)) = \hom(F, \widehat{\mathrm{CFI}}_1(B))$, which proves the lemma. It remains to define Q_η and χ_η and to prove Equation (6).

For each homomorphism $\eta : F \to \widehat{B}$, define the following graphs and map. Define a subgraph F_η of F with vertex set $V(F)$ by retaining an edge $vw \in E(F)$ if and only if η maps vw to an edge of one of the attached copies T_z in $\widehat{B}$. Let Q_η be the simple graph obtained from F by contracting each connected component of F_η to one vertex and replacing parallel edges by single edges; its vertices are the connected components of F_η. Two distinct components are adjacent in Q_η if and only if an edge of F has one endpoint in each component. Every vertex of $\widehat{B}$ lies in exactly one attached copy, and the edges of F_η map into attached copies, so for each connected component K of F_η there is a unique $z \in V(B)$ whose copy of T_z contains $\eta(K)$; we define $\chi_\eta(K) = z$. If $vw \in E(F)$ has $v \in K$ and $w \in L$ for distinct components K, L, then $\eta(v)\eta(w)$ is an edge of B with endpoints $\chi_\eta(K)$ and $\chi_\eta(L)$. Hence $\chi_\eta : Q_\eta \to B$ is a homomorphism.

To prove Equation (6), we fix η and construct a bijection between homomorphisms $f : F \to \widehat{\mathrm{CFI}}_i(B)$ satisfying $\widehat{\pi}_i \circ f = \eta$ and lifts $h : Q_\eta \to \mathrm{CFI}_i(B)$ of χ_η. Given such an f, each connected component K of F_η maps into a single attached copy, since every edge of K projects into some T_z. Let $h(K)$ be the root of this copy. Then $\pi_i \circ h = \chi_\eta$, and edges between components map to edges of $\mathrm{CFI}_i(B)$, so h is a lift of χ_η.

Conversely, let h be a lift of χ_η. For each component K of F_η, a copy of $T_{\chi_\eta(K)}$ is attached at $h(K)$ because $\pi_i(h(K)) = \chi_\eta(K)$. The projection $\widehat{\pi}_i$ maps this copy isomorphically onto the copy of $T_{\chi_\eta(K)}$ in $\widehat{B}$. For each $v \in K$, there is therefore a unique vertex $f(v)$ in the copy rooted



Operation X Operation Y

Figure 2: The operations X and Y applied to a rooted subcubic caterpillar T with root r. New vertices are white, and the new root r' is circled.

at $h(K)$ such that $\widehat{\pi}_i(f(v)) = \eta(v)$. Edges within a component are preserved by this isomorphism. If $vw \in E(F)$ has $v \in K$ and $w \in L$ for distinct components K, L, then $\eta(v)$ and $\eta(w)$ are roots. Hence $f(v) = h(K)$ and $f(w) = h(L)$, which are adjacent because h is a homomorphism. The map $f \mapsto h$ and the map $h \mapsto f$ are inverse to each other, which proves the bijection. Summing over $\eta : F \to \widehat{B}$ proves Equation (6). □

We next choose the attached graphs to obtain simple spectrum (Lemma 10). A *rooted subcubic caterpillar* is a tree obtained from a nonempty path by attaching at most one new leaf to each path vertex and choosing an endpoint of the path as the root. This path is called its *spine*. Thus the maximum degree is at most three and the root degree is at most two; these bounds are used for the degree bound in Lemma 10. The family $\mathcal{T}$ consists of the rooted subcubic caterpillars obtained from K_2 rooted at an endpoint by zero or more applications of the following two operations. For a tree with root r, operation X adds two new vertices r', v and the edges $r'v, vr$, making r' the new root. Operation Y adds a new root r' and a new leaf v, with edges $rr', r'v$ (Figure 2).

We compute the characteristic polynomial of each $T \in \mathcal{T}$ recursively along its construction. The adjacency matrix of a graph H is denoted by A_H. For the root r of T, let $p_T(\lambda) = \det(\lambda I - A_T)$ and $q_T(\lambda) = \det(\lambda I - A_{T-r})$ be the characteristic polynomials of T and $T - r$, respectively. For a nonzero column $\xi = (p, q)^\top \in \mathbb{R}^2$, let $[\xi] = [p : q] \in \mathbb{P}^1(\mathbb{R})$ denote its span, identified with p/q when $q \neq 0$ and with ∞ when $q = 0$. In Lemma 8, the eigenvalue multiplicities of the rooted products depend on each attached tree T only through the point $h_T(\lambda) = [p_T(\lambda) : q_T(\lambda)]$ (see Lemma 7). We show that this point is defined for every real λ, that is, $p_T(\lambda)$ and $q_T(\lambda)$ never vanish together. The starting pair for rooted K_2 is $(p_{K_2}(\lambda), q_{K_2}(\lambda)) = (\lambda^2 - 1, \lambda)$. By cofactor expansion, the operations X and Y transform the column $(p_T(\lambda), q_T(\lambda))^\top$ by multiplication with the respective matrices

$$\mathbf{X}_\lambda = \begin{pmatrix} \lambda^2 - 1 & -\lambda \\ \lambda & -1 \end{pmatrix}, \quad \mathbf{Y}_\lambda = \begin{pmatrix} \lambda^2 - 1 & -\lambda \\ \lambda & 0 \end{pmatrix}. \tag{7}$$

At $\lambda = 0$, the recurrence Equation (7) gives $p_T(0) = (-1)^{|V(T)|/2}$ and $q_T(0) = 0$. These values are used for nonsingularity in Lemma 8. At $\lambda \neq 0$, the starting pair is nonzero and both matrices in Equation (7) are invertible, so they carry nonzero pairs to nonzero pairs. Thus $h_T(\lambda)$ is defined for every real λ.

Lemma 6. *For any finite list of pairs $(\lambda_j, a_j) \in \mathbb{R}^\times \times \mathbb{P}^1(\mathbb{R})$, there is a tree $T \in \mathcal{T}$ with $h_T(\lambda_j) \neq a_j$ for every j.*

Proof. Let ω be a finite composition of the construction operations X, Y, and let $T = \omega(K_2)$. For each j, let $M_{\omega,j}$ be the matrix product obtained by replacing each X by $\mathbf{X}_{\lambda_j}$, each Y by $\mathbf{Y}_{\lambda_j}$, and composition by matrix multiplication. For the empty composition, $M_{\omega,j} = I$. Let $\nu_j = (p_{K_2}(\lambda_j), q_{K_2}(\lambda_j))^\top = (\lambda_j^2 - 1, \lambda_j)^\top$. Then $(p_T(\lambda_j), q_T(\lambda_j))^\top = M_{\omega,j}\nu_j$ by Equation (7). Therefore, it suffices to find a composition ω such that $[M_{\omega,j}\nu_j] \neq a_j$ for every j.

We express the conditions $[M_{\omega,j}\nu_j] \neq a_j$ for all j as the nonvanishing of a linear functional on a tensor product. For each j, define the linear functional $\ell_j : \mathbb{R}^2 \to \mathbb{R}$ by

$$\ell_j(p, q) = \begin{cases} p - a_j q, & a_j \in \mathbb{R}, \\ q, & a_j = \infty. \end{cases} \tag{8}$$

Let $\nu = \bigotimes_j \nu_j$, $\ell = \bigotimes_j \ell_j$, $\mathbf{X} = \bigotimes_j \mathbf{X}_{\lambda_j}$ and $\mathbf{Y} = \bigotimes_j \mathbf{Y}_{\lambda_j}$, where $\otimes$ is the tensor product. Define M_ω in the same way using $\mathbf{X}, \mathbf{Y}$. Multiplication of tensor products gives $M_\omega = \bigotimes_j M_{\omega,j}$. Since ℓ is a tensor product of functionals,

$$\ell(M_\omega \nu) = \prod_j \ell_j(M_{\omega,j}\nu_j), \tag{9}$$

and each factor is nonzero exactly when $h_T(\lambda_j) \neq a_j$. Thus the avoidance conditions hold exactly when $\ell(M_\omega \nu) \neq 0$.

It suffices to find a finite product W of the matrices $\mathbf{X}^{\pm 1}, \mathbf{Y}^{\pm 1}$ with $\ell(W\nu) \neq 0$. Since every $\lambda_j \neq 0$, the matrices $\mathbf{X}, \mathbf{Y}$ are invertible. By the Cayley–Hamilton theorem, their inverses are polynomials in $\mathbf{X}, \mathbf{Y}$, respectively. Substituting these polynomial expressions into W and expanding gives a finite sum $W = \sum_i c_i M_{\omega_i}$, where $c_i \in \mathbb{R}$ and each ω_i is a composition of the construction operations X, Y. Consequently, $\ell(W\nu) = \sum_i c_i \ell(M_{\omega_i}\nu) \neq 0$ implies $\ell(M_{\omega_i}\nu) \neq 0$ for some i. By Equation (9), the tree $T = \omega_i(K_2)$ then satisfies $h_T(\lambda_j) \neq a_j$ for every j.

We now construct such a product W. For $\lambda \neq 0$, the matrices in Equation (7) act on $\mathbb{P}^1(\mathbb{R})$ by multiplication of column representatives. Let $D_\lambda = \mathbf{X}_\lambda^{-1}\mathbf{Y}_\lambda$. On $\mathbb{P}^1(\mathbb{R})$, D_λ acts by $t \mapsto \lambda^{-2}t + \lambda^{-1}$. The starting value is $h_{K_2}(\lambda) = \lambda - \lambda^{-1} = \mathbf{Y}_\lambda \infty$. Since $\mathbf{Y}_\lambda^{-1}\infty = 0$, we have $\mathbf{Y}_\lambda^{-2} h_{K_2}(\lambda) = 0$. Thus $D_\lambda^m \mathbf{Y}_\lambda^{-2}$ sends the starting value to $\lambda^{-1}\sum_{k=0}^{m-1}\lambda^{-2k}$. These values are strictly monotone in m for every real $\lambda \neq 0$, so each a_j is attained for at most one m at $\lambda = \lambda_j$. Choose a positive integer m such that $[D_{\lambda_j}^m \mathbf{Y}_{\lambda_j}^{-2}\nu_j] \neq a_j$ for every j. Then $W = (\mathbf{X}^{-1}\mathbf{Y})^m \mathbf{Y}^{-2}$ satisfies $\ell(W\nu) = \prod_j \ell_j(D_{\lambda_j}^m \mathbf{Y}_{\lambda_j}^{-2}\nu_j) \neq 0$, as required. □

For a base graph with edge weights, the rooted product retains the original edge weights, and the edges of each T_z have weight one.

Lemma 7. *Let B be a graph with real edge weights. For each $z \in V(B)$, let $T_z \in \mathcal{T}$, and let $\widehat{B}$ be the rooted product of B with $(T_z)_{z\in V(B)}$. For $\lambda \in \mathbb{R}$, define*

$$K^\lambda = \operatorname{diag}(p_{T_z}(\lambda))_{z\in V(B)} - A_B \operatorname{diag}(q_{T_z}(\lambda))_{z\in V(B)}. \tag{10}$$

Then

$$\dim\ker(\lambda I - A_{\widehat{B}}) = \dim\ker K^\lambda. \tag{11}$$

Proof. Fix $\lambda \in \mathbb{R}$ and put $p_z = p_{T_z}(\lambda)$ and $q_z = q_{T_z}(\lambda)$. Let $J = \{z \in V(B) : q_z \neq 0\}$ and $J^c = V(B) \setminus J$, and define

$$\Sigma = \operatorname{diag}(p_z/q_z)_{z\in J} - A_B[J, J], \tag{12}$$

where $A_B[J, J]$ is the principal submatrix indexed by J. We will show that both sides of Equation (11) equal $\dim\ker\Sigma$. When $J = \varnothing$, we have $\dim\ker\Sigma = 0$.

We first consider K^λ. For $z \in J^c$, we have $q_z = 0$ and $p_z \neq 0$, so column z has the single nonzero entry p_z in row z. Ordering J before J^c therefore gives

$$K^\lambda = \begin{pmatrix} K^\lambda[J, J] & 0 \\ K^\lambda[J^c, J] & \operatorname{diag}(p_z)_{z\in J^c} \end{pmatrix}. \tag{13}$$

Since the lower-right block is invertible, $\dim\ker K^\lambda = \dim\ker K^\lambda[J, J]$. Also $K^\lambda[J, J] = \Sigma \operatorname{diag}(q_z)_{z\in J}$ with every $q_z \neq 0$, so $\dim\ker K^\lambda = \dim\ker\Sigma$.

It remains to prove $\dim\ker(\lambda I - A_{\widehat{B}}) = \dim\ker\Sigma$. Let $\widehat{B[J]}$ be the rooted product of the induced base graph $B[J]$ with $(T_z)_{z\in J}$. We first prove the deletion identity

$$\dim\ker(\lambda I - A_{\widehat{B}}) = \dim\ker(\lambda I - A_{\widehat{B[J]}}). \tag{14}$$

To prove Equation (14), we delete from $\widehat{B}$ the whole copies T_z with $z \in J^c$, including their roots, one at a time. At each step, let L be the remaining induced subgraph of $\widehat{B}$. For the copy T_z to be removed, put $D_z = \lambda I - A_{T_z}$ and $R_z = \lambda I - A_{L-V(T_z)}$. Let e_z be the root coordinate vector in $\mathbb{R}^{V(T_z)}$, and let b_z be the vector of edge weights from the root to $V(L) \setminus V(T_z)$. Ordering $V(T_z)$ first gives

$$\lambda I - A_L = \begin{pmatrix} D_z & -e_z b_z^\top \\ -b_z e_z^\top & R_z \end{pmatrix}. \tag{15}$$

Here D_z is invertible because $\det D_z = p_z \neq 0$. Cramer's rule gives $e_z^\top D_z^{-1} e_z = q_z/p_z = 0$. Thus the Schur complement of D_z in Equation (15) is

$$R_z - b_z(e_z^\top D_z^{-1} e_z) b_z^\top = R_z - \frac{q_z}{p_z} b_z b_z^\top = R_z. \tag{16}$$

Since D_z is invertible, $\dim\ker(\lambda I - A_L) = \dim\ker R_z$. The remaining matrix is exactly $R_z = \lambda I - A_{L-V(T_z)}$, so the same calculation applies after each deletion. Deleting all copies indexed by J^c therefore proves Equation (14).

It remains to show that $\dim\ker(\lambda I - A_{\widehat{B[J]}}) = \dim\ker\Sigma$. For $z \in J$, put $E_z = \lambda I - A_{T_z - r_z}$, and let c_z be the vector of edge weights from r_z to the non-root vertices of T_z. Let $E = \bigoplus_{z\in J} E_z$, and let C have rows indexed by these non-root vertices and columns indexed by J. Column z of C equals c_z on $V(T_z) \setminus \{r_z\}$ and is zero elsewhere. Ordering all non-root vertices before the roots gives

$$\lambda I - A_{\widehat{B[J]}} = \begin{pmatrix} E & -C \\ -C^\top & \lambda I - A_B[J,J] \end{pmatrix}. \tag{17}$$

Every E_z is invertible because $\det E_z = q_z \neq 0$. The determinant formula for the Schur complement of E_z in $\lambda I - A_{T_z}$ gives $p_z = q_z(\lambda - c_z^\top E_z^{-1} c_z)$. Since distinct columns of C are supported on distinct blocks of E, the Schur complement of E in Equation (17) is

$$\lambda I - A_B[J,J] - C^\top E^{-1} C = \operatorname{diag}(p_z/q_z)_{z\in J} - A_B[J,J] = \Sigma. \tag{18}$$

Since E is invertible, Equations (14) and (18) give

$$\dim\ker(\lambda I - A_{\widehat{B}}) = \dim\ker\Sigma = \dim\ker K^\lambda, \tag{19}$$

which proves Equation (11). For unweighted rooted products, the determinant identity $\det(\lambda I - A_{\widehat{B}}) = \det K^\lambda$ is [9, Theorem 2.1]. □

Lemma 8. *Let Z be a finite set, and let $B_1, \dots, B_s$ be connected graphs with nonzero real edge weights and pairwise distinct nonempty vertex sets $Z_i = V(B_i) \subseteq Z$. There is a family $(T_z)_{z\in Z}$ of trees in $\mathcal{T}$ such that the rooted products of B_i with $(T_z)_{z\in Z_i}$ are nonsingular and have simple, pairwise disjoint spectra.*

Proof. Put $d = |Z|$, and let A_{B_i} be the weighted adjacency matrix of B_i. For $\alpha, \beta \in \mathbb{C}^Z$, let

$$K_i(\alpha,\beta) = \operatorname{diag}(\alpha_z)_{z\in Z_i} - A_{B_i} \operatorname{diag}(\beta_z)_{z\in Z_i}. \tag{20}$$

For a projective point $a = ([\alpha_z : \beta_z])_{z\in Z}$, rescaling each homogeneous representative multiplies $K_i(\alpha,\beta)$ on the right by an invertible diagonal matrix. Thus $\dim\ker K_i(\alpha,\beta)$ depends only on a. Fix trees $T_z \in \mathcal{T}$, and write $\widehat{B}_i$ for the rooted product of B_i with $(T_z)_{z\in Z_i}$. For $\lambda \in \mathbb{R}$, let $K_i^\lambda = K_i((p_{T_z}(\lambda))_{z\in Z}, (q_{T_z}(\lambda))_{z\in Z})$.

We define closed subsets $\mathcal{V}_i^{(k)}$ for $k \in \{1,2\}$ and $\mathcal{E}$ of $(\mathbb{P}^1(\mathbb{C}))^Z$ by

$$\mathcal{V}_i^{(k)} = \{([\alpha_z : \beta_z])_{z\in Z} : \dim\ker K_i(\alpha,\beta) \geq k\}, \quad \mathcal{E} = \bigcup_i \mathcal{V}_i^{(2)} \cup \bigcup_{i\neq i'} (\mathcal{V}_i^{(1)} \cap \mathcal{V}_{i'}^{(1)}). \tag{21}$$

These conditions are independent of the homogeneous representatives. By Lemma 7, for $\lambda \in \mathbb{R}$ and $a = (h_{T_z}(\lambda))_{z\in Z}$, the number λ is an eigenvalue of $\widehat{B}_i$ of multiplicity at least k exactly when $a \in \mathcal{V}_i^{(k)}$. It is an eigenvalue of both $\widehat{B}_i$ and $\widehat{B}_{i'}$ exactly when $a \in \mathcal{V}_i^{(1)} \cap \mathcal{V}_{i'}^{(1)}$. At $\lambda = 0$, the matrix K_i^0 is invertible because $q_{T_z}(0) = 0$ and $p_{T_z}(0) \neq 0$. Thus the products $\widehat{B}_i$ are nonsingular. Their spectra are simple and pairwise disjoint exactly when

$$(h_{T_z}(\lambda))_{z\in Z} \notin \mathcal{E} \quad \text{for every } \lambda \in \mathbb{R}^\times. \tag{22}$$

We next prove $\dim\mathcal{E} \leq d-2$ over $\mathbb{C}$. The polynomial $\det(\operatorname{diag}(t) - A_{B_i})$ in the variables $(t_z)_{z\in Z_i}$ is irreducible because B_i is connected [1, Lemma 4.1]. Substituting $t_z = \alpha_z/\beta_z$ and clearing denominators gives the determinant of the matrix in Equation (20), as a polynomial in

the homogeneous representatives (α_z, β_z). Its projective zero set is $\mathcal{V}_i^{(1)}$, so $\mathcal{V}_i^{(1)}$ is an irreducible hypersurface. For each $z \in Z$, let $a^{(z)} \in (\mathbb{P}^1(\mathbb{C}))^Z$ have coordinate $[0:1]$ at z and $[1:0]$ elsewhere. Using these homogeneous coordinates, $\dim \ker K_i(\alpha, \beta)$ is one if $z \in Z_i$ and zero otherwise. For $i \neq i'$, choose z in the symmetric difference of Z_i and $Z_{i'}$; then $a^{(z)}$ lies in exactly one of $\mathcal{V}_i^{(1)}$ and $\mathcal{V}_{i'}^{(1)}$. Thus the $\mathcal{V}_i^{(1)}$ are distinct, so each intersection $\mathcal{V}_i^{(1)} \cap \mathcal{V}_{i'}^{(1)}$ has dimension at most $d-2$. For each i and each $z \in Z_i$, the point $a^{(z)}$ lies in $\mathcal{V}_i^{(1)} \setminus \mathcal{V}_i^{(2)}$. Thus $\mathcal{V}_i^{(2)}$ is a proper closed subset of $\mathcal{V}_i^{(1)}$ and also has dimension at most $d-2$. The union defining $\mathcal{E}$ therefore has dimension at most $d-2$.

Since $\mathbb{C}^\times \times \mathcal{E}$ has dimension at most $d-1$, we can choose d trees to satisfy Equation (22) as follows. We enumerate Z as $\{1, \ldots, d\}$. For fixed trees $T_1, \ldots, T_j$, let P_j be the set of pairs (λ, a) with $\lambda \in \mathbb{R}^\times$, $a \in \mathcal{E} \cap (\mathbb{P}^1(\mathbb{R}))^Z$, and $a_z = h_{T_z}(\lambda)$ for all $z \leq j$. Let Ω_j be the Zariski closure of P_j in $\mathbb{C}^\times \times \mathcal{E}$. Then $\dim \Omega_0 \leq d-1$. We choose T_{j+1} so that Ω_{j+1} contains no irreducible component of Ω_j. By definition, $P_{j+1} \subseteq P_j$, so $\Omega_{j+1} \subseteq \Omega_j$. Every pair $(\lambda, a) \in P_{j+1}$ satisfies $a_{j+1} = h_{T_{j+1}}(\lambda)$. In homogeneous coordinates this is equivalent to

$$\alpha_{j+1} q_{T_{j+1}}(\lambda) - \beta_{j+1} p_{T_{j+1}}(\lambda) = 0. \tag{23}$$

This polynomial equation also holds on the Zariski closure Ω_{j+1}. It therefore suffices to choose T_{j+1} so that Equation (23) fails at one point of every irreducible component of Ω_j.

For each irreducible component $\mathcal{W}$ of Ω_j, we choose a pair $(\lambda_\mathcal{W}, a_\mathcal{W}) \in P_j \cap \mathcal{W}$ that lies in no other irreducible component of Ω_j. Such a pair exists, because otherwise P_j would lie in the closed union of the other components, contradicting the definition of Ω_j as the Zariski closure of P_j. The pairs $(\lambda_\mathcal{W}, (a_\mathcal{W})_{j+1})$, one for each irreducible component $\mathcal{W}$ of Ω_j, form a finite list in $\mathbb{R}^\times \times \mathbb{P}^1(\mathbb{R})$. By Lemma 6 there is therefore a tree $T_{j+1} \in \mathcal{T}$ with $h_{T_{j+1}}(\lambda_\mathcal{W}) \neq (a_\mathcal{W})_{j+1}$ for every $\mathcal{W}$. Thus Equation (23) fails at $(\lambda_\mathcal{W}, a_\mathcal{W})$ in every irreducible component $\mathcal{W}$ of Ω_j. Thus Ω_{j+1} contains no irreducible component of Ω_j, so $\dim \Omega_{j+1} < \dim \Omega_j$. If Ω_j is empty, we choose the remaining trees arbitrarily. Since $\dim \Omega_0 \leq d-1$ and the dimension strictly decreases whenever Ω_j is nonempty, $\Omega_d = \varnothing$, so $P_d = \varnothing$. Hence Equation (22) holds, and the lemma follows. □

Lemma 9. *Let Γ be a connected cubic graph, and let S be its subdivision with one new vertex on each edge. Choose the distinguished vertex v_* of $\mathrm{CFI}_1(S)$ in $V(\Gamma)$. Fix trees $(T_z)_{z \in V(S)}$ in $\mathcal{T}$ and form the CFI-compatible rooted products $\widehat{\mathrm{CFI}}_i(S)$ for $i \in \{0,1\}$. The subdivision has vertex set $V(S) = V(\Gamma) \sqcup E(\Gamma)$, where the new vertex on the edge e is denoted by e. Let S_0 be S with every edge weighted $\sqrt{2}$, and for $e = vw \in E(\Gamma)$ let S_e be the path with edges ve, ew and the same edge weights. Write $\widehat{S}_0$ and $\widehat{S}_e$ for their rooted products with the corresponding trees T_z. For each $i \in \{0,1\}$, the adjacency matrix of $\widehat{\mathrm{CFI}}_i(S)$ is orthogonally similar to*

$$A_{\widehat{S}_0} \oplus \bigoplus_{e \in E(\Gamma)} A_{\widehat{S}_e}. \tag{24}$$

Proof. We prove the decomposition by specifying the corresponding subspaces and orthonormal bases. Fix $i \in \{0,1\}$. For $v \in V(\Gamma)$, the fibre over v is indexed by $\Xi_i(v)$, which consists of the four vectors in $\mathbb{F}_2^{E_S(v)}$ of the prescribed parity. For a subdivision vertex e, the fibre over e is indexed by $\Xi_i(e) = \{(0,0),(1,1)\}$; we write b for (b,b). The vertices of $\widehat{\mathrm{CFI}}_i(S)$ are indexed by (z, ξ, x), with $z \in V(S)$, $\xi \in \Xi_i(z)$ and $x \in V(T_z)$. The coordinate (z, ξ, r_z) denotes the original CFI vertex (z, ξ), where r_z is the root of T_z.

On $\Xi_i(v)$, we use the basis $\phi_{v,0}(\xi) = 1/2$ and $\phi_{v,e}(\xi) = (-1)^{\xi_e}/2$ for the three edges e incident with v. On $\Xi_i(e)$, we use the basis $\psi_{e,0}(b) = 1/\sqrt{2}$ and $\psi_{e,1}(b) = (-1)^b/\sqrt{2}$. These bases are orthonormal under either parity constraint. On $\Xi_i(v)$, any two incident coordinates range exactly once over $\mathbb{F}_2^2$, because parity determines the third. Thus $\sum_{\xi \in \Xi_i(v)} \phi_{v,e}(\xi) = 0$, $\langle \phi_{v,e}, \phi_{v,e'} \rangle = 0$ for distinct incident edges e, e', and $\|\phi_{v,e}\| = 1$. The functions $\psi_{e,0}, \psi_{e,1}$ are orthonormal by the same two-point calculation.

For $z \in V(S)$, $g \in \mathbb{R}^{\Xi_i(z)}$ and $x \in V(T_z)$, let $g[x] \in \mathbb{R}^{V(\widehat{\mathrm{CFI}}_i(S))}$ be the function with value $g(\xi)$ at (z, ξ, x) for each $\xi \in \Xi_i(z)$ and zero elsewhere. The functions $g[x]$ obtained from all basis vectors g above and all tree positions x form an orthonormal basis of $\mathbb{R}^{V(\widehat{\mathrm{CFI}}_i(S))}$. We define subspaces corresponding to $\widehat{S}_0$ and the graphs $\widehat{S}_e$. Let U_0 be the subspace of functions constant in ξ at each fixed (z, x). The subspace U_0 is spanned by $\phi_{v,0}[x]$ for $v \in V(\Gamma)$ and $x \in V(T_v)$, together with

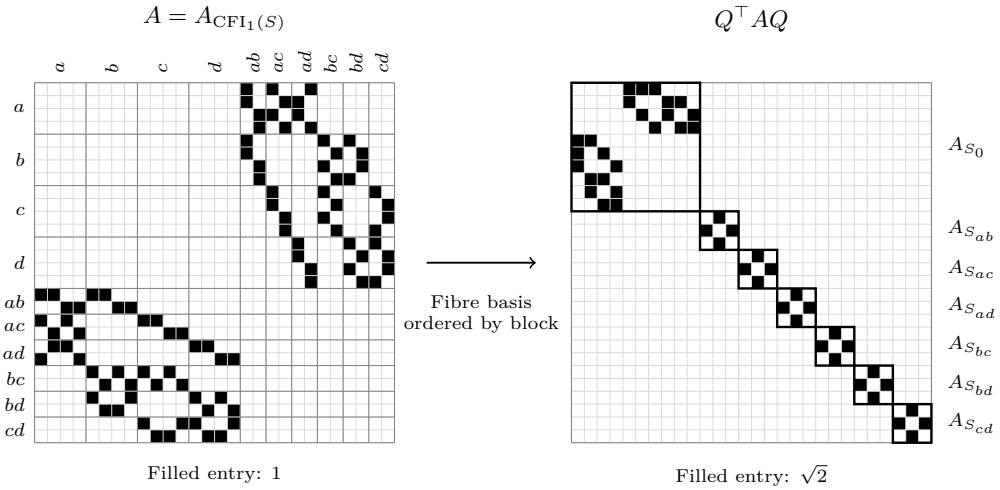


Figure 3: Block decomposition for the subdivision S of $\Gamma = K_4$ on vertices a, b, c, d, with tree attachments omitted. The left matrix groups the 28 CFI vertices by fibre and uses odd parity at a. Within fibres, the bit vectors are in lexicographic order, with incident-edge coordinates ordered by their labels. The columns of Q are the fibre basis vectors from the proof, ordered first by the ten constant directions and then by the three sign directions over v, e, w for each edge $e = vw$ of K_4. The right matrix has one 10×10 block $A_{S_0} = \sqrt{2}A_S$ and six 3×3 blocks A_{S_e}; unfilled entries are zero. Both CFI parities give the same right matrix.

$\psi_{e,0}[x]$ for $e \in E(\Gamma)$ and $x \in V(T_e)$. For each $e = vw$, let U_e be spanned by $\phi_{v,e}[x]$, $\psi_{e,1}[x]$ and $\phi_{w,e}[x]$, with x ranging over $V(T_v), V(T_e), V(T_w)$, respectively. These spanning sets partition the orthonormal basis. For each subspace, we send each of its listed basis vectors $g[x]$ to the standard coordinate vector indexed by (z, x), where $g \in \mathbb{R}^{\Xi_i(z)}$. This gives the isometric identifications

$$U_0 \cong \mathbb{R}^{V(\widehat{S}_0)}, \quad U_e \cong \mathbb{R}^{V(\widehat{S}_e)} \text{ for } e \in E(\Gamma). \tag{25}$$

We compute the adjacency matrix in this basis, starting with tree edges. For $z \in V(S)$, write $(g_\kappa)_{\kappa=1}^{|\Xi_i(z)|}$ for its fibre basis. Let Q_z be the matrix whose κth column is g_κ, so $(Q_z)_{\xi\kappa} = g_\kappa(\xi)$ for $\xi \in \Xi_i(z)$. The subgraph of $\widehat{\mathrm{CFI}}_i(S)$ induced by the vertices (z, ξ, x) with $\xi \in \Xi_i(z)$ and $x \in V(T_z)$ is the disjoint union of $|\Xi_i(z)|$ copies of T_z. Its adjacency matrix in coordinates (ξ, x) is $I \otimes A_{T_z}$, because the same tree is attached at every vertex of the fibre. The same basis change at every tree position is $Q_z \otimes I$, so

$$(Q_z \otimes I)^\top (I \otimes A_{T_z})(Q_z \otimes I) = (Q_z^\top Q_z) \otimes A_{T_z} = I \otimes A_{T_z}. \tag{26}$$

Thus the adjacency matrix of this induced subgraph has one block A_{T_z} on each set $\{g_\kappa[x] : x \in V(T_z)\}$, with zero entries between these sets. Each set lies in one of the subspaces U_0, U_e.

For an incidence $v \in e$, the root vertices (v, ξ, r_v) and (e, b, r_e) are adjacent exactly when $\xi_e = b$. Let $R_{v,e}$ be the matrix $R_{v,e}(\xi, b) = \delta_{\xi_e, b}$ of this adjacency. It satisfies

$$R_{v,e}\psi_{e,0} = \sqrt{2}\phi_{v,0}, \quad R_{v,e}\psi_{e,1} = \sqrt{2}\phi_{v,e}, \tag{27}$$

and, by orthonormality, $R_{v,e}^\top$ sends all other vectors in the chosen basis of $\mathbb{R}^{\Xi_i(v)}$ to zero. Thus the matrix of $R_{v,e}$ in these fibre bases has entry $\sqrt{2}$ at the row–column pairs $(\phi_{v,0}, \psi_{e,0})$ and $(\phi_{v,e}, \psi_{e,1})$, and all other entries are zero. Hence, in the stated bases of U_0 and U_e, the submatrices indexed by root coordinates are A_{S_0} and A_{S_e}, respectively. Figure 3 shows this decomposition for $\Gamma = K_4$.

Combining these root-coordinate matrices with the tree blocks in Equation (26) shows that the subspaces U_0 and U_e are invariant and that the adjacency matrix of $\widehat{\mathrm{CFI}}_i(S)$ has the block diagonal form Equation (24) in the stated bases. □

Lemma 10. *Let Γ be a connected cubic graph, and let S be its subdivision with one new vertex on each edge. There is a family $(T_z)_{z\in V(S)}$ of trees in $\mathcal{T}$ such that the CFI-compatible rooted products $\widehat{\mathrm{CFI}}_0(S)$ and $\widehat{\mathrm{CFI}}_1(S)$ are non-isomorphic, bipartite, cospectral, nonsingular graphs with simple spectrum and maximum degree at most six.*

Proof. Since the isomorphism type of $\mathrm{CFI}_1(S)$ is independent of the distinguished vertex v_*, we choose $v_* \in V(\Gamma)$. Let S_0 and S_e be the weighted graphs from Lemma 9. We apply Lemma 8 to these graphs, with $Z = V(S)$. They are connected and have nonzero edge weights. Their vertex sets $V(S)$ and $\{v, e, w\}$ for $e = vw \in E(\Gamma)$ are pairwise distinct. By that lemma, there is one tree $T_z \in \mathcal{T}$ for each $z \in V(S)$ such that the rooted products $\widehat{S}_0$ and $\widehat{S}_e$ are nonsingular and have simple, pairwise disjoint spectra. The block matrix in Equation (24) therefore has simple spectrum and is nonsingular. By Lemma 9, the adjacency matrices of $\widehat{\mathrm{CFI}}_0(S)$ and $\widehat{\mathrm{CFI}}_1(S)$ are both orthogonally similar to this matrix, so the two rooted products are cospectral, nonsingular and have simple spectrum.

The graphs $\widehat{\mathrm{CFI}}_0(S)$ and $\widehat{\mathrm{CFI}}_1(S)$ are non-isomorphic because their maximal subgraphs of minimum degree at least two are the non-isomorphic graphs $\mathrm{CFI}_0(S)$ and $\mathrm{CFI}_1(S)$. Both rooted products are bipartite, since each CFI graph is bipartite and attaching trees preserves bipartiteness. The CFI graphs have maximum degree four, and each attached tree is subcubic with root degree at most two, so the rooted products have maximum degree at most six. □

Proof of Lemma 4. Lemma 10 needs the subdivision of a connected cubic graph as its base, so we first replace B by such a base S that contains B as a minor. Let t be an integer with $t \geq \max\{4, |V(B)|\}$, so that B is a minor of K_t and the cycles below have length at least three. Let Γ be the connected cubic graph obtained from K_t by replacing each vertex with a cycle of length $t-1$ and attaching its $t-1$ incident edges to distinct cycle vertices. Let S be the subdivision of Γ with one new vertex on each edge. Contracting one edge at each subdivision vertex of S gives Γ, and contracting the replacement cycles of Γ gives K_t, so B is a minor of S. Hence $\mathrm{CFI}_0(S) \equiv_{\mathcal{C}} \mathrm{CFI}_1(S)$ follows from the assumption $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$ by [15, Corollary 5.7]. Using the trees from Lemma 10, let $G = \widehat{\mathrm{CFI}}_0(S)$ and $G' = \widehat{\mathrm{CFI}}_1(S)$. That lemma shows that G and G' are non-isomorphic, bipartite, cospectral and nonsingular, with simple spectrum and maximum degree at most six. By Lemma 5, $G \equiv_{\mathcal{C}} G'$. □

Remark 11. For the families in Proposition 3, arbitrarily large clique minors suffice for completeness. It is natural to ask whether every class complete on simple spectrum must contain graphs with arbitrarily large clique minors. The weak Roberson conjecture states that for every proper minor-closed graph class $\mathcal{C}$ there are non-isomorphic graphs $G \equiv_{\mathcal{C}} G'$ [15, Conjecture 5 and Theorem 8.4]. An affirmative answer to the question above implies this conjecture, because a proper minor-closed class $\mathcal{C}$ excludes some K_t, so, under an affirmative answer, there are non-isomorphic graphs $G \equiv_{\mathcal{C}} G'$ with simple spectrum. By Lemma 4, non-isomorphic graphs $G \equiv_{\mathcal{C}} G'$ with simple spectrum exist for every minor-closed class $\mathcal{C}$ that admits a connected graph B with $\mathrm{CFI}_0(B) \equiv_{\mathcal{C}} \mathrm{CFI}_1(B)$.

3 Proof of Theorem 2

We first identify exactly what subcubic homomorphism counts determine on arbitrary graphs. For a graph G with adjacency matrix A_G, let $W_G = \operatorname{im} A_G \subseteq \mathbb{R}^{V(G)}$. For $d \in \{0, 1, 2, 3\}$ and $f_1, \ldots, f_d \in \mathbb{R}^{V(G)}$, we define

$$\langle f_1, \ldots, f_d \rangle_G = \sum_{x \in V(G)} \prod_{i=1}^{d} f_i(x). \tag{28}$$

The empty product is 1, so $\langle\rangle_G = |V(G)|$. The cases with one, two and three arguments are the linear form, the standard inner product and the symmetric trilinear form, respectively. We equip W_G with the restricted inner product.

We use finite multigraphs, allowing loops and parallel edges; a loop contributes two to the degree. For a multigraph H and a tuple $\mathbf{m} = (m_e)_{e\in E(H)}$ of positive integers, $H^{\mathbf{m}}$ is the subdivision that

replaces each edge e by a path of length m_e. For a subdividing path with q_e internal vertices, its length is $m_e = q_e + 1$. When every $m_e \geq 3$, this subdivision is simple, even if H has loops or parallel edges.

Subdivision counts have an expansion in the spectral projectors of the target graph. For a graph G with adjacency matrix A_G, let Λ_G be the set of nonzero eigenvalues of A_G. For $\lambda \in \Lambda_G$, E_λ denotes the λ-eigenspace and P_λ its orthogonal projector. For $\lambda \notin \Lambda_G$, including $\lambda = 0$, we set $E_\lambda = \{0\}$ and $P_\lambda = 0$. We add a graph subscript, as in $E_{\lambda,G}$ and $P_{\lambda,G}$, when needed. For a multigraph H and an edge labelling $\ell : E(H) \to \mathbb{R}^\times$, the *labelled coefficient* is defined by

$$c_G(H,\ell) = \sum_{\varphi : V(H) \to V(G)} \prod_{e=vw \in E(H)} P_{\ell(e)}(\varphi(v), \varphi(w)). \tag{29}$$

For positive lengths making $H^{\mathbf{m}}$ simple,

$$\hom(H^{\mathbf{m}}, G) = \sum_{\ell : E(H) \to \Lambda_G} c_G(H,\ell) \prod_{e \in E(H)} \ell(e)^{m_e}. \tag{30}$$

Indeed, for a path of length m_e with endpoint images fixed at x, y, summing over the images of its internal vertices gives $A_G^{m_e}(x,y) = \sum_{\lambda \in \Lambda_G} \lambda^{m_e} P_\lambda(x,y)$. Multiplying out these expansions over all edges gives Equation (30).

Lemma 12. *For two graphs G, G' and a multigraph H, $\hom(H^{\mathbf{m}}, G) = \hom(H^{\mathbf{m}}, G')$ for every simple subdivision $H^{\mathbf{m}}$ if and only if $c_G(H,\ell) = c_{G'}(H,\ell)$ for every edge labelling $\ell : E(H) \to \mathbb{R}^\times$.*

Proof. Equality of the labelled coefficients gives equality of the counts by Equation (30): since the coefficients vanish whenever a label lies outside the target's nonzero spectrum, both sums may be taken over $\ell : E(H) \to \Lambda_G \cup \Lambda_{G'}$. Conversely, suppose the counts agree for every simple subdivision. The functions $m \mapsto \lambda^m$ for $\lambda \in \Lambda_G \cup \Lambda_{G'}$ are linearly independent on the integers $m \geq 3$, by the Vandermonde determinant. Comparing the coefficients of $\prod_e \ell(e)^{m_e}$ in Equation (30) for G and for G', one variable $m_e \geq 3$ at a time, gives $c_G(H,\ell) = c_{G'}(H,\ell)$ for every ℓ. □

We call G and G' *adjacency-image equivalent*, written $G \sim_{\mathrm{im}} G'$, if $|V(G)| = |V(G')|$ and there is a bijective linear map $U : W_G \to W_{G'}$ such that

$$U(A_G f) = A_{G'}(Uf), \quad \langle Uf, Ug \rangle_{G'} = \langle f, g \rangle_G, \quad \langle Uf, Ug, Uh \rangle_{G'} = \langle f, g, h \rangle_G \tag{31}$$

for every $f, g, h \in W_G$.

A *tensor* on a Euclidean space is a multilinear form, and its *coefficients* in a basis are its values on tuples of basis vectors. A *closed tensor network* $\mathcal{N}$ is a finite multigraph with a tensor at each vertex and one tensor index for each incident edge. Loops and parallel edges are allowed; a loop uses two indices at its vertex. In an orthonormal basis, its value $\mathrm{val}(\mathcal{N})$ is obtained by multiplying tensor coefficients and summing over the indices paired along edges. The same orthogonal change of basis at every vertex leaves this value unchanged. We use the following result, called the converse of the real orthogonal Holant theorem.

Theorem ([17, Theorem 9], specialised form)**.** *Let G, G' be graphs with $\dim W_G = \dim W_{G'}$. For a finite multigraph $\mathcal{N}$ whose vertices all have degree two or three, let $\mathrm{val}_G(\mathcal{N})$ be the value of the closed tensor network on W_G with underlying multigraph $\mathcal{N}$ that assigns the adjacency form $(f,g) \mapsto \langle f, A_G g \rangle_G$ to each degree-two vertex and the trilinear form $\langle \cdot, \cdot, \cdot \rangle_G$ to each degree-three vertex. Let $\mathrm{val}_{G'}(\mathcal{N})$ be the value of the same construction with G' in place of G. Then the equality*

$$\mathrm{val}_G(\mathcal{N}) = \mathrm{val}_{G'}(\mathcal{N}) \tag{32}$$

holds for every such multigraph $\mathcal{N}$ if and only if there is a bijective linear isometry $U : W_G \to W_{G'}$ satisfying $U(A_G f) = A_{G'}(Uf)$ and $\langle Uf, Ug, Uh \rangle_{G'} = \langle f, g, h \rangle_G$ for all $f, g, h \in W_G$.

Lemma 13. *For any finite simple graphs G, G', the following are equivalent:*

(a) $G \equiv_{\mathcal{D}_3} G'$;

(b) $G \sim_{\mathrm{im}} G'$*;*

(c) $|V(G)| = |V(G')|$ *and* $c_G(H,\ell) = c_{G'}(H,\ell)$ *for every connected cubic multigraph* H *and every edge labelling* $\ell : E(H) \to \mathbb{R}^\times$*.*

Proof. We first prove (b)⇒(a). Suppose $G \sim_{\mathrm{im}} G'$, witnessed by U. Let $(u_i)_{i\in I^*}$ be an orthonormal eigenbasis of W_G, with $A_G u_i = \lambda_i u_i$. Then (Uu_i) is an orthonormal eigenbasis of $W_{G'}$, since U is bijective, preserves the inner product, and satisfies $A_{G'}Uu_i = UA_Gu_i = \lambda_i Uu_i$.

We first show that U also preserves the linear form:

$$\langle Uf\rangle_{G'} = \langle f\rangle_G \tag{33}$$

for every $f \in W_G$. Let $\mathbf{1}_G \in \mathbb{R}^{V(G)}$ be the constant-one function. Expanding A_G^2 in the eigenbasis gives $\sum_i ((A_G u_i)(x))^2 = A_G^2(x,x)$. Since G is simple and undirected, $A_G^2(x,x) = \deg_G(x) = (A_G\mathbf{1}_G)(x)$. Multiplying by $h(x)$ and summing over x gives, for every $h \in W_G$,

$$\sum_i \langle h, A_G u_i, A_G u_i\rangle_G = \langle h, A_G\mathbf{1}_G\rangle_G = \langle A_G h\rangle_G. \tag{34}$$

Now let $f \in W_G$. Since $A_G|_{W_G}$ has only nonzero eigenvalues, it is invertible, so $f = A_G g$ for some $g \in W_G$. Equation (34) also holds for G' with the eigenbasis (Uu_i). Hence, since $A_{G'}U = UA_G$ and U preserves the trilinear form,

$$\begin{aligned}\langle Uf\rangle_{G'} &= \langle A_{G'}Ug\rangle_{G'} = \sum_i \langle Ug, A_{G'}Uu_i, A_{G'}Uu_i\rangle_{G'} \\ &= \sum_i \langle Ug, UA_Gu_i, UA_Gu_i\rangle_{G'} = \sum_i \langle g, A_Gu_i, A_Gu_i\rangle_G = \langle A_Gg\rangle_G = \langle f\rangle_G,\end{aligned} \tag{35}$$

which is Equation (33).

Fix a subcubic graph F. Expanding each adjacency factor in the homomorphism count as $A_G(x,y) = \sum_{i\in I^*} \lambda_i u_i(x)u_i(y)$ and summing first over the images of the vertices of F gives

$$\hom(F,G) = \sum_{\iota : E(F)\to I^*} \left(\prod_{e\in E(F)} \lambda_{\iota(e)}\right) \prod_{v\in V(F)} \left\langle u_{\iota(e)} : e \ni v\right\rangle_G. \tag{36}$$

The expansion for G' in the basis $(Uu_i)_{i\in I^*}$ has the same eigenvalue factors. The corresponding vertex factors agree by the assumption $G \sim_{\mathrm{im}} G'$ for degrees $0, 2, 3$, and by Equation (33) for degree 1. Hence $\hom(F,G) = \hom(F,G')$.

We next prove (a)⇒(c). Suppose $G \equiv_{\mathcal{D}_3} G'$. The count from K_1 gives $|V(G)| = |V(G')|$. Every simple subdivision of a connected cubic multigraph H is subcubic, so its homomorphism counts into G and G' agree. Lemma 12 gives $c_G(H,\ell) = c_{G'}(H,\ell)$ for every nonzero edge labelling ℓ.

We now prove (c)⇒(b). Suppose (c) holds. We first show that G and G' have the same nonzero eigenvalues with the same multiplicities. Let Θ be the cubic multigraph with vertices a, b and three parallel edges. We fix an order of these edges and write $c_G(\Theta;\lambda,\mu,\nu)$ for its coefficient when these edges receive the labels λ, μ, ν, respectively. For every $\lambda \in \Lambda_G \cup \Lambda_{G'}$,

$$\begin{aligned}\sum_{\mu,\nu\in\Lambda_G\cup\Lambda_{G'}} \mu\nu c_G(\Theta;\lambda,\mu,\nu) &= \sum_{x,y\in V(G)} P_{\lambda,G}(x,y)A_G(x,y)^2 \\ &= \sum_{x,y\in V(G)} P_{\lambda,G}(x,y)A_G(x,y) \\ &= \operatorname{tr}(P_{\lambda,G}A_G) = \lambda \dim E_{\lambda,G}.\end{aligned} \tag{37}$$

The first equality follows from Equation (29) and $A_G = \sum_{\mu\in\Lambda_G\cup\Lambda_{G'}} \mu P_{\mu,G}$; the second uses that G is simple, so every adjacency entry is 0 or 1. The same identity holds for G'. The assumed equality of the labelled coefficients equates the left sides, so division by $\lambda \neq 0$ gives $\dim E_{\lambda,G} = \dim E_{\lambda,G'}$. Every nonzero real number outside $\Lambda_G \cup \Lambda_{G'}$ has multiplicity zero for both graphs. The adjacency

images W_G and $W_{G'}$ therefore have the same dimension. If this dimension is zero, both graphs are edgeless, so the unique linear map $W_G \to W_{G'}$ witnesses $G \sim_{\mathrm{im}} G'$.

Since $\dim W_G = \dim W_{G'}$ and $|V(G)| = |V(G')|$, [17, Theorem 9] in the specialised form above gives $G \sim_{\mathrm{im}} G'$ if Equation (32) holds for every such $\mathcal{N}$. Since network values multiply over connected components, it is enough to prove $\mathrm{val}_G(\mathcal{N}) = \mathrm{val}_{G'}(\mathcal{N})$ for every connected $\mathcal{N}$.

Fix a connected multigraph $\mathcal{N}$ of this kind. Either every vertex of $\mathcal{N}$ has degree two, or $\mathcal{N}$ has a vertex of degree three; we treat the two cases in turn. If every vertex of $\mathcal{N}$ has degree two, then $\mathcal{N}$ is a cycle. The cyclic contraction of its $|V(\mathcal{N})|$ adjacency forms gives

$$\mathrm{val}_G(\mathcal{N}) = \mathrm{tr}\big((A_G|_{W_G})^{|V(\mathcal{N})|}\big). \tag{38}$$

Since $A_G|_{W_G}$ and $A_{G'}|_{W_{G'}}$ have the same spectrum, their $|V(\mathcal{N})|$th powers have equal traces. Thus $\mathrm{val}_G(\mathcal{N}) = \mathrm{val}_{G'}(\mathcal{N})$.

Now suppose $\mathcal{N}$ has a vertex of degree three. Let $\overline{\mathcal{N}}$ be the cubic multigraph obtained from $\mathcal{N}$ by repeatedly removing a degree-two vertex and replacing its two incident edges with a single edge between its two neighbours, or a loop if they coincide. Each edge e of $\overline{\mathcal{N}}$ replaces a path in $\mathcal{N}$; let $q_e \geq 0$ be the number of degree-two vertices on that path. Figure 4 illustrates the network, its cubic multigraph, and a simple subdivision. We claim that

$$\mathrm{val}_G(\mathcal{N}) = \sum_{\ell : E(\overline{\mathcal{N}}) \to \Lambda_G} \left(\prod_{e \in E(\overline{\mathcal{N}})} \ell(e)^{q_e} \right) c_G(\overline{\mathcal{N}}, \ell). \tag{39}$$

Contracting the q_e adjacency forms along the path represented by e gives the bilinear form $(f, g) \mapsto \langle f, (A_G|_{W_G})^{q_e} g \rangle_G$ on W_G, including the inner product when $q_e = 0$. We evaluate the remaining contraction in an orthonormal eigenbasis $(u_i)_{i \in I^*}$ of W_G, with $A_G u_i = \lambda_i u_i$. The bilinear form for e has coefficients $\lambda_i^{q_e} \delta_{ij}$. Expanding each remaining trilinear coefficient as $\langle u_i, u_j, u_k \rangle_G = \sum_x u_i(x) u_j(x) u_k(x)$ and interchanging the sums gives

$$\mathrm{val}_G(\mathcal{N}) = \sum_{\varphi : V(\overline{\mathcal{N}}) \to V(G)} \prod_{e = vw \in E(\overline{\mathcal{N}})} \left(\sum_i \lambda_i^{q_e} u_i(\varphi(v)) u_i(\varphi(w)) \right). \tag{40}$$

Grouping each inner sum by its eigenvalue gives $\sum_{\lambda \in \Lambda_G} \lambda^{q_e} P_{\lambda, G}(\varphi(v), \varphi(w))$. Multiplying out the edge factors now gives Equation (39), by Equation (29).

By (c), $c_G(\overline{\mathcal{N}}, \ell) = c_{G'}(\overline{\mathcal{N}}, \ell)$ for every nonzero edge labelling ℓ. Since $\Lambda_G = \Lambda_{G'}$, Equation (39) gives

$$\mathrm{val}_G(\mathcal{N}) = \sum_{\ell : E(\overline{\mathcal{N}}) \to \Lambda_G} \left(\prod_e \ell(e)^{q_e} \right) c_{G'}(\overline{\mathcal{N}}, \ell) = \mathrm{val}_{G'}(\mathcal{N}). \tag{41}$$

Thus Equation (32) holds. By [17, Theorem 9] there is a bijective linear map U that preserves the inner product, intertwines the adjacency operators and preserves the trilinear forms; since also $|V(G)| = |V(G')|$, $G \sim_{\mathrm{im}} G'$. □

We next show that adjacency-image equivalence implies isomorphism for graphs of nullity at most one. For a graph G, we extend an orthonormal eigenbasis of W_G to an orthonormal eigenbasis $(u_i)_{i \in I}$ of $\mathbb{R}^{V(G)}$. Let λ_i be the eigenvalue of u_i for each $i \in I$, and let $I^* = \{i \in I : \lambda_i \neq 0\}$. Let γ_1 and γ_3 be the coefficient arrays of the linear and trilinear forms in this eigenbasis, given by

$$\gamma_1(i) = \langle u_i \rangle_G, \quad \gamma_3(i, j, k) = \langle u_i, u_j, u_k \rangle_G \tag{42}$$

for $i, j, k \in I$. For $x \in V(G)$, let $r_x = (u_i(x))_{i \in I} \in \mathbb{R}^I$ be the row of x. These rows form an orthonormal basis of $\mathbb{R}^I$. For $r \in \mathbb{R}^I$, let $r^{\otimes 3}$ denote the array with entries $r(i) r(j) r(k)$ for $i, j, k \in I$. The rows satisfy

$$\gamma_3 = \sum_{x \in V(G)} r_x^{\otimes 3}, \quad A_G(x, y) = \sum_{i \in I} \lambda_i r_x(i) r_y(i). \tag{43}$$

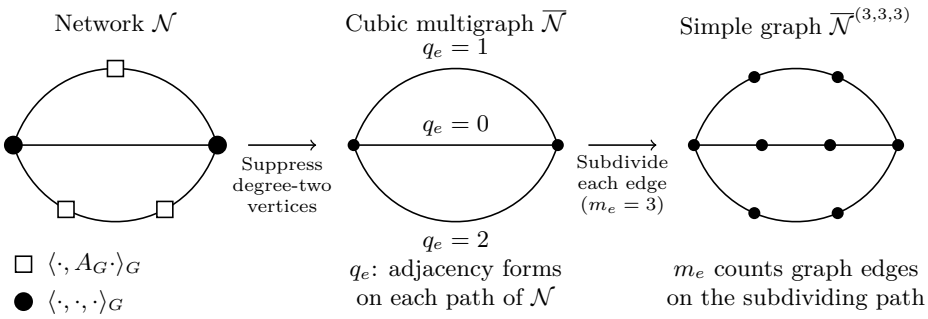


Figure 4: A network, its cubic multigraph, and one simple subdivision. The path represented by e has q_e adjacency forms and $q_e + 1$ edges in $\mathcal{N}$; a direct edge has $q_e = 0$. The subdivision on the right is chosen independently of the network paths, with $m_e = 3$ on every edge. The expansions of $\mathrm{val}_G(\mathcal{N})$ and $\hom(\overline{\mathcal{N}}^{\mathbf{m}}, G)$ use the same coefficients $c_G(\overline{\mathcal{N}}, \ell)$, with weights $\prod_e \ell(e)^{q_e}$ and $\prod_e \ell(e)^{m_e}$, respectively. Varying the lengths $m_e \geq 3$ determines these coefficients and hence the network value for every choice of $q_e \geq 0$.

We write γ_3^* for the restriction of γ_3 to indices in I^*. In the eigenbasis $(u_i)_{i \in I^*}$ of W_G, the form $\langle \cdot, \cdot, \cdot \rangle_G$ has coefficient array γ_3^*. When G has nullity one, we index the basis so that $I = I^* \sqcup \{0\}$ and u_0 spans $\ker A_G$. In particular, $\gamma_1(0) = \sum_{x \in V(G)} u_0(x)$. For such a graph, let $\tilde{r}_x = (u_i(x))_{i \in I^*}$ be its reduced row at x. Since $\tilde{r}_x$ is r_x with its kernel coordinate omitted, orthonormality of the rows gives

$$\langle \tilde{r}_x, \tilde{r}_y \rangle = \delta_{xy} - u_0(x) u_0(y). \tag{44}$$

The coefficients of γ_3 outside γ_3^* are the $\gamma_3(i, j, k)$ with at least one index equal to 0.

If G has nullity one and $|\operatorname{supp} u_0| > 2$, then

$$N_G(x) \neq N_G(y) \quad \text{for distinct } x, y \in \operatorname{supp} u_0. \tag{45}$$

Indeed, for the standard coordinate vectors $e_x, e_y \in \mathbb{R}^{V(G)}$, $e_x - e_y$ has support of size two, so it is not a scalar multiple of u_0. Thus $A_G(e_x - e_y) \neq 0$, which gives Equation (45).

The final assertion of the next lemma is a special case of uniqueness of orthogonal tensor decompositions [18, Theorem 3.2].

Lemma 14. *Let X, I be finite sets, and let $R \in \mathbb{R}^{X \times I}$ be a matrix with orthonormal columns. Write $r_x \in \mathbb{R}^I$ for its row indexed by $x \in X$, and define $\kappa : \mathbb{R}^I \to \mathbb{R}$ by $\kappa(\xi) = \sum_{x \in X} \langle r_x, \xi \rangle^3$. Then*

$$\{\xi : \|\xi\| = 1,\ \kappa(\xi) = 1\} = \{r_x : \|r_x\| = 1\}. \tag{46}$$

In particular, if $(r_x)_{x \in X}$ and $(r'_y)_{y \in Y}$ are orthonormal bases of the same Euclidean space and $\sum_x r_x^{\otimes 3} = \sum_y (r'_y)^{\otimes 3}$, then there is a bijection $\pi : X \to Y$ with $r'_{\pi(x)} = r_x$ for every $x \in X$.

Proof. Let ξ be a unit vector and put $c = R\xi$. Since the columns of R are orthonormal, we have $\sum_x c_x^2 = 1$. Hence

$$1 - \kappa(\xi) = \sum_x c_x^2 (1 - c_x) \geq 0. \tag{47}$$

Equality holds exactly when $c = e_x$ for some x, where e_x is the standard coordinate vector. Then $\xi = R^\top c = r_x$. Conversely, if $\|r_x\| = 1$, then Rr_x is a unit vector whose x-coordinate is 1, so $Rr_x = e_x$ and $\kappa(r_x) = 1$. This proves Equation (46). The final assertion follows because equal cubic tensors identify the same set of unit rows. □

Lemma 15. *Let G, G' have nullity one and orthonormal eigenbases $(u_i)_{i \in I}$ and $(u'_i)_{i \in I}$, respectively, with the same corresponding eigenvalues λ_i. Index both bases by $I = I^* \sqcup \{0\}$, with kernel vectors u_0, u'_0, and write $\tilde{r}_x, \tilde{r}'_y \in \mathbb{R}^{I^*}$ for the reduced rows. Suppose that $|\operatorname{supp} u_0| = |\operatorname{supp} u'_0| = 4$ and $\sum_x u_0(x) = \sum_y u'_0(y) = 0$. Suppose also that the reduced rows outside these supports agree as multisets and $\sum_{x \in \operatorname{supp} u_0} \tilde{r}_x = \sum_{y \in \operatorname{supp} u'_0} \tilde{r}'_y$. Then the reduced rows on the two supports agree as multisets.*

Proof. The rows outside each support are orthonormal by Equation (44). We relabel the vertices of G' outside $\operatorname{supp} u_0'$ so that $\tilde{r}'_z = \tilde{r}_z$ for every $z \notin \operatorname{supp} u_0$. We first show that $\operatorname{supp} u_0$ admits at least two distinct partitions into pairs on which u_0 sums to zero. For $w \in V(G)$, let $N_0(w) = N_G(w) \cap \operatorname{supp} u_0$. Then $\sum_{x \in N_0(w)} u_0(x) = (A_G u_0)(w) = 0$. If a nonempty proper subset T of $\operatorname{supp} u_0$ satisfies $\sum_{x \in T} u_0(x) = 0$, then $|T| = 2$ and $\sum_{x \in \operatorname{supp} u_0 \setminus T} u_0(x) = 0$, since the four support entries are nonzero and sum to zero. Choose distinct $x, y \in \operatorname{supp} u_0$. By Equation (45), there is a vertex w_1 for which $N_0(w_1)$ contains exactly one of x, y. Thus $N_0(w_1)$ is nonempty and proper, and $\{N_0(w_1), \operatorname{supp} u_0 \setminus N_0(w_1)\}$ is a first partition into zero-sum pairs. The two vertices of $N_0(w_1)$ also have different neighbourhoods, so there is a vertex w_2 for which $N_0(w_2)$ contains exactly one of them. Again $N_0(w_2)$ is nonempty and proper, so $\{N_0(w_2), \operatorname{supp} u_0 \setminus N_0(w_2)\}$ is a partition into zero-sum pairs. It differs from the first partition because its two pairs separate the vertices of $N_0(w_1)$.

We label $\operatorname{supp} u_0 = \{a, b, c, d\}$ so that u_0 sums to zero on each pair in the two partitions $\{\{a, c\}, \{b, d\}\}$ and $\{\{a, d\}, \{b, c\}\}$. Since u_0 sums to zero on each of $\{a, c\}$, $\{a, d\}$ and $\{b, c\}$, we have $u_0(c) = u_0(d) = -u_0(a) = -u_0(b)$. Since u_0 is a unit vector, a choice of sign gives

$$u_0(a) = u_0(b) = \frac{1}{2}, \quad u_0(c) = u_0(d) = -\frac{1}{2}. \tag{48}$$

Every $N_0(w)$ is then empty, equal to $\operatorname{supp} u_0$, or one of $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$. Since $\sum_y u_0'(y) = 0$ and $|\operatorname{supp} u_0'| = 4$, the same argument gives the nonzero entries $(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$ for u_0', up to permutation and a choice of sign.

We now show that $\{\tilde{r}_x : x \in \operatorname{supp} u_0\}$ and $\{\tilde{r}'_y : y \in \operatorname{supp} u_0'\}$ have the same centre and span the same affine plane. We choose the sign of u_0' and label $\operatorname{supp} u_0' = \{a', b', c', d'\}$ so that $(u_0'(a'), u_0'(b'), u_0'(c'), u_0'(d')) = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$. Let $\sigma = \tilde{r}_a + \tilde{r}_b$, $f = (\tilde{r}_a - \tilde{r}_b)/\sqrt{2}$ and $g = (\tilde{r}_c - \tilde{r}_d)/\sqrt{2}$. Let σ', f', g' be given by the formulas for σ, f, g with primed vertices and rows. By Equation (44) and Equation (48), the vectors σ, f, g together with $(\tilde{r}_z)_{z \in V(G) \setminus \operatorname{supp} u_0}$ form an orthonormal basis of $\mathbb{R}^{I^*}$. Since u_0 is orthogonal to every u_i with $i \in I^*$, we have $\sum_x u_0(x) \tilde{r}_x = 0$. Substituting Equation (48) gives $\tilde{r}_a + \tilde{r}_b = \tilde{r}_c + \tilde{r}_d$, so $\sigma = \tilde{r}_c + \tilde{r}_d$. Hence

$$\tilde{r}_a = \frac{1}{2}\sigma + \frac{1}{\sqrt{2}} f, \quad \tilde{r}_b = \frac{1}{2}\sigma - \frac{1}{\sqrt{2}} f, \quad \tilde{r}_c = \frac{1}{2}\sigma + \frac{1}{\sqrt{2}} g, \quad \tilde{r}_d = \frac{1}{2}\sigma - \frac{1}{\sqrt{2}} g. \tag{49}$$

The centres of $\{\tilde{r}_a, \tilde{r}_b, \tilde{r}_c, \tilde{r}_d\}$ and $\{\tilde{r}'_{a'}, \tilde{r}'_{b'}, \tilde{r}'_{c'}, \tilde{r}'_{d'}\}$ are their averages $\sigma/2$ and $\sigma'/2$. By the assumed equality of the support-row sums,

$$\frac{1}{2}\sigma' = \frac{1}{4} \sum_{y \in \operatorname{supp} u_0'} \tilde{r}'_y = \frac{1}{4} \sum_{x \in \operatorname{supp} u_0} \tilde{r}_x = \frac{1}{2}\sigma. \tag{50}$$

Since $\sigma' = \sigma$ and $\tilde{r}'_z = \tilde{r}_z$ for $z \notin \operatorname{supp} u_0$, the orthonormal bases $\{\sigma, f, g\} \cup \{\tilde{r}_z : z \notin \operatorname{supp} u_0\}$ and $\{\sigma', f', g'\} \cup \{\tilde{r}'_z : z \notin \operatorname{supp} u_0'\}$ give

$$\operatorname{span}(f, g) = \operatorname{span}\big(\{\tilde{r}_z : z \notin \operatorname{supp} u_0\} \cup \{\sigma\}\big)^{\perp} = \operatorname{span}(f', g'). \tag{51}$$

Thus both sets of four vectors have affine span $\sigma/2 + \operatorname{span}(f, g)$.

We now use the entries of $A_{G'}$ to show that $f' \in \{\pm f, \pm g\}$. Let $\alpha, \beta \in \mathbb{R}$ be the coefficients in $f' = \alpha f + \beta g$. Since (f, g) is orthonormal and f' is a unit vector, $\alpha^2 + \beta^2 = 1$. For each $z \in V(G) \setminus \operatorname{supp} u_0$, let $\Delta_z^f = A_G(a, z) - A_G(b, z)$ and $\Delta_z^g = A_G(c, z) - A_G(d, z)$. By Equations (43) and (49),

$$\begin{aligned} \sqrt{2} f^{\top} \operatorname{diag}(\lambda_i)_{i \in I^*} \tilde{r}_z &= \Delta_z^f, \\ \sqrt{2} g^{\top} \operatorname{diag}(\lambda_i)_{i \in I^*} \tilde{r}_z &= \Delta_z^g, \\ \sigma^{\top} \operatorname{diag}(\lambda_i)_{i \in I^*} \tilde{r}_z &= A_G(a, z) + A_G(b, z). \end{aligned} \tag{52}$$

Since $\sigma' = \sigma$ and $f' = \alpha f + \beta g$, we have $\tilde{r}'_{a'} = \frac{1}{2}\sigma + \frac{1}{\sqrt{2}}(\alpha f + \beta g)$ by Equation (49) for G'. Substituting this formula for $\tilde{r}'_{a'}$ and $\tilde{r}'_z = \tilde{r}_z$ into the second identity in Equation (43) for G', and using Equation (52), gives

$$A_{G'}(a', z) = \frac{1}{2}\big(A_G(a, z) + A_G(b, z)\big) + \frac{1}{2}(\alpha \Delta_z^f + \beta \Delta_z^g) \in \{0, 1\}. \tag{53}$$

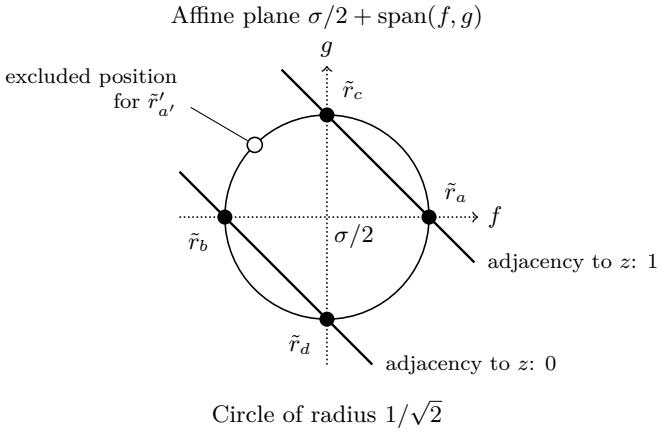


Figure 5: Geometric interpretation of the calculation giving $\alpha\beta = 0$. The figure assumes that $z \in V(G) \setminus \operatorname{supp} u_0$ is adjacent to a, c and not to b, d; the other cases of Equation (54) are symmetric. For a candidate $\tilde{r}'_{a'} = \sigma/2 + (\alpha f + \beta g)/\sqrt{2}$, the circle gives $\alpha^2 + \beta^2 = 1$. By Equation (53), the two lines have equations $\alpha + \beta = 1$ and $\alpha + \beta = -1$. Squaring either equation and subtracting the circle equation gives $2\alpha\beta = 0$. The hollow point is on the circle but off both lines, so Equation (53) excludes it.

We have $\alpha\beta = 0$ whenever a vertex $z \notin \operatorname{supp} u_0$ satisfies

$$A_G(a, z) + A_G(b, z) = A_G(c, z) + A_G(d, z) = 1. \tag{54}$$

Indeed, then $\Delta^f_z, \Delta^g_z \in \{\pm 1\}$ and $A_G(a, z) + A_G(b, z) = 1$, so Equation (53) gives $\alpha\Delta^f_z + \beta\Delta^g_z = \pm 1$. Squaring gives $1 = \alpha^2 + \beta^2 + 2\alpha\beta\Delta^f_z\Delta^g_z = 1 + 2\alpha\beta\Delta^f_z\Delta^g_z$, so $\alpha\beta = 0$. Figure 5 gives a geometric interpretation of this calculation.

We show that Equation (54) holds for some $z \in V(G) \setminus \operatorname{supp} u_0$. Otherwise, the list of possible sets $N_0(z)$ implies $N_0(z) \in \{\varnothing, \operatorname{supp} u_0\}$ for every $z \notin \operatorname{supp} u_0$. For $w \in \operatorname{supp} u_0$, we have $w \notin N_0(w)$ because G has no loops. Thus a nonempty $N_0(w)$ must be one of $\{a, c\}$, $\{a, d\}$, $\{b, c\}$ and $\{b, d\}$. Every nonisolated vertex in the graph induced by $\operatorname{supp} u_0$ therefore has exactly two neighbours, one in $\{a, b\}$ and one in $\{c, d\}$. A nonisolated vertex in $\{a, b\}$ is adjacent to the other vertex of $\{a, b\}$; the same holds for $\{c, d\}$. If the induced graph has an edge, it therefore also has an edge between $\{a, b\}$ and $\{c, d\}$. Each of these two sets then contains a nonisolated vertex and hence both its vertices are nonisolated. All four vertices have degree two, so the induced graph is a cycle of length four. Thus either the induced graph is edgeless, or it is a four-cycle. There are distinct $x, y \in \operatorname{supp} u_0$ with $N_0(x) = N_0(y)$: any two vertices in the edgeless case, or opposite vertices in the four-cycle case. Also $N_G(x) \setminus \operatorname{supp} u_0 = N_G(y) \setminus \operatorname{supp} u_0$, because every vertex outside the support is adjacent to all four support vertices or to none. Hence $N_G(x) = N_G(y)$, contradicting $N_G(x) \neq N_G(y)$ for distinct $x, y \in \operatorname{supp} u_0$.

A vertex $z \notin \operatorname{supp} u_0$ satisfying Equation (54) exists, so $\alpha\beta = 0$. Together with $\alpha^2 + \beta^2 = 1$, this gives $(\alpha, \beta) \in \{(1, 0), (-1, 0), (0, 1), (0, -1)\}$, hence $f' \in \{\pm f, \pm g\}$. Since (f', g') is orthonormal in $\operatorname{span}(f, g)$, we have $\{\pm f', \pm g'\} = \{\pm f, \pm g\}$. Together with $\sigma' = \sigma$, Equation (49) now gives

$$\{\tilde{r}_a, \tilde{r}_b, \tilde{r}_c, \tilde{r}_d\} = \{\tilde{r}'_{a'}, \tilde{r}'_{b'}, \tilde{r}'_{c'}, \tilde{r}'_{d'}\}. \tag{55}$$

The four vectors on each side are distinct by Equation (49), so they agree as multisets. □

Lemma 16. *Let G have nullity at most one, and let G' be a graph. If $G \sim_{\text{im}} G'$, then $G \cong G'$.*

Proof. We fix a bijective linear map $U : W_G \to W_{G'}$ witnessing $G \sim_{\text{im}} G'$. The graphs G and G' have the same number of vertices because $G \sim_{\text{im}} G'$. The map U intertwines $A_G|_{W_G}$ with $A_{G'}|_{W_{G'}}$, so their nonzero eigenvalues agree with multiplicities. Thus their zero multiplicities also agree, and both graphs have the same nullity, at most one. We fix an orthonormal eigenbasis $(u_i)_{i \in I}$ of $\mathbb{R}^{V(G)}$. Let $(u'_i)_{i \in I}$ be an orthonormal eigenbasis of $\mathbb{R}^{V(G')}$ with $u'_i = Uu_i$ for every $i \in I^*$. Then u_i and

u'_i have the same eigenvalue λ_i for every $i \in I$. We use primes for quantities associated with G'. By Equation (31) and Equation (33), for $i, j, k \in I^*$,

$$\begin{aligned} \gamma'_1(i) &= \langle Uu_i \rangle_{G'} = \langle u_i \rangle_G = \gamma_1(i), \\ \gamma'_3(i,j,k) &= \langle Uu_i, Uu_j, Uu_k \rangle_{G'} = \langle u_i, u_j, u_k \rangle_G = \gamma_3(i,j,k). \end{aligned} \tag{56}$$

For $x \in V(G)$ and $y \in V(G')$, let $\tilde{r}_x = (u_i(x))_{i \in I^*}$ and $\tilde{r}'_y = (u'_i(y))_{i \in I^*}$ be the reduced rows. A bijection $\pi : V(G) \to V(G')$ with $\tilde{r}'_{\pi(x)} = \tilde{r}_x$ is a graph isomorphism. The zero-eigenvalue terms in the second identity of Equation (43) vanish, so for $x, y \in V(G)$ we have

$$A_{G'}(\pi(x), \pi(y)) = \sum_{i \in I^*} \lambda_i \tilde{r}'_{\pi(x)}(i) \tilde{r}'_{\pi(y)}(i) = \sum_{i \in I^*} \lambda_i \tilde{r}_x(i) \tilde{r}_y(i) = A_G(x, y). \tag{57}$$

If $\gamma_3 = \gamma'_3$, then by Lemma 14 there is a bijection π with $r'_{\pi(x)} = r_x$, hence $\tilde{r}'_{\pi(x)} = \tilde{r}_x$. If the graphs are nonsingular, then $I^* = I$, so Equation (56) already gives $\gamma_3 = \gamma'_3$ and hence $G \cong G'$.

We therefore assume that both graphs have nullity one. We index (u_i) and (u'_i) so that $I = I^* \sqcup \{0\}$, with unit kernel vectors u_0, u'_0. Once $(u_i)_{i \in I^*}$ and $(u'_i)_{i \in I^*}$ are fixed, the only remaining choices are the signs of u_0 and u'_0. We first compare the coefficients involving index 0, which are not compared in Equation (56). By symmetry of γ_3, the *missing coefficients* are $\gamma_1(0)$, $\gamma_3(0,0,0)$, $\gamma_3(0,0,k)$ and $\gamma_3(0,j,k)$ for $j, k \in I^*$. We seek a choice of signs for u_0, u'_0 for which these coefficients agree with their primed counterparts. If these coefficients agree, then Equation (56) gives $\gamma_3 = \gamma'_3$, and Lemma 14 gives a vertex bijection π satisfying $\tilde{r}'_{\pi(v)} = \tilde{r}_v$ for every $v \in V(G)$. Replacing u_0 by $-u_0$ negates $\gamma_1(0)$, $\gamma_3(0,0,0)$ and $\gamma_3(0,j,k)$, while leaving $\gamma_3(0,0,k)$ unchanged. Replacing u'_0 by $-u'_0$ has the same effect on the primed coefficients.

The arrays γ_1, γ_3 satisfy the following identities for all $i, j, k, l \in I$ because the vectors r_x form an orthonormal basis:

$$\sum_{m \in I} \gamma_1(m)^2 = |V(G)|, \tag{58}$$

$$\sum_{m \in I} \gamma_1(m) \gamma_3(m,j,k) = \delta_{jk}, \tag{59}$$

$$\sum_{m \in I} \gamma_3(j,m,m) = \gamma_1(j), \tag{60}$$

$$\sum_{m \in I} \gamma_3(i,j,m) \gamma_3(m,k,l) = \sum_{x \in V(G)} u_i(x) u_j(x) u_k(x) u_l(x). \tag{61}$$

Separating the term $m = 0$ in Equation (60) gives, for $k \in I^*$,

$$\begin{aligned} \gamma_3(0,0,k) &= \gamma_1(k) - \sum_{i \in I^*} \gamma_3(k,i,i), \\ \gamma_3(0,0,0) &= \gamma_1(0) - \sum_{i \in I^*} \gamma_3(0,i,i). \end{aligned} \tag{62}$$

The first identity involves only coefficients at indices in I^* on the right-hand side, so $\gamma'_3(0,0,k) = \gamma_3(0,0,k)$ by Equation (56). The second determines $\gamma_3(0,0,0)$ from $\gamma_1(0)$ and the coefficients $\gamma_3(0,j,k)$. Thus agreement of all missing coefficients reduces to agreement of $\gamma_1(0)$ and $\gamma_3(0,j,k)$ with their primed counterparts.

For the case analysis, let C_0 be the symmetric $I^* \times I^*$ matrix defined by

$$\mathsf{C}_0(j,k) = \gamma_3(0,j,k). \tag{63}$$

Separating the term $m = 0$ in Equation (58) gives $\gamma_1(0)^2 = |V(G)| - \sum_{i \in I^*} \gamma_1(i)^2$, so $\gamma_1(0)^2 = (\gamma'_1(0))^2$ by Equation (56). We divide the remaining proof into $\gamma_1(0) \neq 0$ (Case A) and $\gamma_1(0) = 0$ (Case B). Within Case B, we distinguish $\operatorname{rank} \mathsf{C}_0 \geq 3$ (Case B1) from $\operatorname{rank} \mathsf{C}_0 \leq 2$ (Case B2). In Cases A and B1, we show that a choice of sign for u'_0 gives $\gamma'_1(0) = \gamma_1(0)$ and $\mathsf{C}'_0 = \mathsf{C}_0$. In Case B2, we use the graph structure to show that u_0 has support of size two or four. In both subcases, we show directly that $(\tilde{r}_x)_{x \in V(G)}$ and $(\tilde{r}'_y)_{y \in V(G')}$ agree as multisets.

Case A: $\gamma_1(0) \neq 0$. We choose the sign of u_0' so that $\gamma_1'(0) = \gamma_1(0)$. Separating the term $m = 0$ in Equation (59) gives

$$\gamma_1(0)\mathsf{C}_0(j,k) = \delta_{jk} - \sum_{i \in I^*} \gamma_1(i)\gamma_3(i,j,k). \tag{64}$$

The right-hand side for G' equals the right-hand side for G by Equation (56), and dividing by $\gamma_1(0) = \gamma_1'(0) \neq 0$ gives $\mathsf{C}_0' = \mathsf{C}_0$. Hence $\gamma_3 = \gamma_3'$.

Case B: $\gamma_1(0) = 0$. Then $\gamma_1'(0) = 0$, since $(\gamma_1'(0))^2 = \gamma_1(0)^2$. Here Equation (59) no longer determines C_0.

Case B1: $\operatorname{rank} \mathsf{C}_0 \geq 3$. We show that $\mathsf{C}_0' = \pm\mathsf{C}_0$ by comparing their 2×2 minors. For $i, j, k, l \in I^*$ let $Q(i, j; k, l)$ be the sum of the terms with $m \neq 0$ in Equation (61),

$$Q(i,j;k,l) = \sum_{m \in I^*} \gamma_3(i,j,m)\gamma_3(m,k,l), \tag{65}$$

which involves only γ_3^*. Separating the term $m = 0$ in Equation (61) gives

$$\mathsf{C}_0(i,j)\mathsf{C}_0(k,l) + Q(i,j;k,l) = \sum_{x \in V(G)} u_i(x)u_j(x)u_k(x)u_l(x). \tag{66}$$

The right-hand side is symmetric in (i, j, k, l), so subtracting from Equation (66) the same identity with j and k exchanged gives

$$\mathsf{C}_0(i,j)\mathsf{C}_0(k,l) - \mathsf{C}_0(i,k)\mathsf{C}_0(j,l) = Q(i,k;j,l) - Q(i,j;k,l). \tag{67}$$

The left-hand side is the minor of C_0 with rows i, l and columns j, k. The right-hand side involves only γ_3^*, which G and G' share by Equation (56). Hence each 2×2 minor of C_0' equals the minor of C_0 with the same rows and columns. Thus the exterior squares of C_0 and C_0' are equal. Since $\operatorname{rank} \mathsf{C}_0 \geq 3$, [10, Corollary 4.1] gives $\mathsf{C}_0' = \pm\mathsf{C}_0$. We choose the sign of u_0' so that $\mathsf{C}_0' = \mathsf{C}_0$. Hence $\gamma_3 = \gamma_3'$.

Case B2: $\operatorname{rank} \mathsf{C}_0 \leq 2$. We show that $\{\tilde{r}_z : z \notin \operatorname{supp} u_0\} = \{\tilde{r}_z' : z \notin \operatorname{supp} u_0'\}$. Let R be the matrix with rows $\tilde{r}_x$. Its columns are orthonormal, and Equation (44) gives $\|\tilde{r}_x\|^2 = 1 - u_0(x)^2$. Thus $\tilde{r}_x$ is a unit vector exactly when $x \notin \operatorname{supp} u_0$. The same holds for $\tilde{r}_y'$ and $\operatorname{supp} u_0'$. Let κ and κ' be the cubic forms with coefficient arrays γ_3^* and $(\gamma_3')^*$, respectively, so

$$\kappa(\xi) = \sum_x \langle \tilde{r}_x, \xi \rangle^3 = \sum_y \langle \tilde{r}_y', \xi \rangle^3 = \kappa'(\xi) \tag{68}$$

by Equation (56). By Lemma 14 and Equation (68),

$$\{\tilde{r}_z : z \notin \operatorname{supp} u_0\} = \{\xi : \|\xi\| = 1,\ \kappa(\xi) = 1\} = \{\tilde{r}_z' : z \notin \operatorname{supp} u_0'\}. \tag{69}$$

Each of these row families is orthonormal by Equation (44), so its members are distinct. Since G and G' have the same number of vertices, Equation (69) also gives $|\operatorname{supp} u_0'| = |\operatorname{supp} u_0|$. We relabel the vertices of G' outside $\operatorname{supp} u_0'$ so that $\tilde{r}_z' = \tilde{r}_z$ for every $z \notin \operatorname{supp} u_0$.

It remains to compare $(\tilde{r}_x)_{x \in \operatorname{supp} u_0}$ and $(\tilde{r}_y')_{y \in \operatorname{supp} u_0'}$ as multisets. Their sums agree: by Equation (56) and Equation (69),

$$\sum_{x \in \operatorname{supp} u_0} \tilde{r}_x = (\gamma_1(i))_{i \in I^*} - \sum_{z \notin \operatorname{supp} u_0} \tilde{r}_z = \sum_{y \in \operatorname{supp} u_0'} \tilde{r}_y'. \tag{70}$$

We claim that

$$|\operatorname{supp} u_0| \in \{2, 4\}. \tag{71}$$

Since $\mathsf{C}_0 = R^\top \operatorname{diag}(u_0) R$ and $\operatorname{rank} R = |V(G)| - 1$, Sylvester's rank inequality gives $|\operatorname{supp} u_0| \leq \operatorname{rank} \mathsf{C}_0 + 2 \leq 4$. The support has at least two vertices because $u_0 \neq 0$ and $\gamma_1(0) = 0$. Suppose it has three vertices. For every nonempty proper subset T of $\operatorname{supp} u_0$, we have $\sum_{x \in T} u_0(x) \neq 0$, since the three entries are nonzero and their sum is zero. The equation $(A_G u_0)(w) = \sum_{x \in N_G(w)} u_0(x) = 0$ therefore forces every vertex w to be adjacent to all three support vertices or to none. Thus $N_G(x) = N_G(y)$ for any two distinct support vertices x, y, contradicting Equation (45).

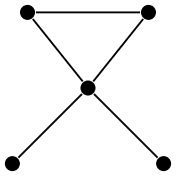


Figure 6: The cricket graph B.

Case B2 with $|\operatorname{supp} u_0| = 2$. Write $\operatorname{supp} u_0 = \{x, y\}$ and $\operatorname{supp} u'_0 = \{x', y'\}$. Since $u_0(x) = -u_0(y) \neq 0$, we have $0 = \langle u_0, u_i\rangle_G = u_0(x)(u_i(x) - u_i(y))$ for all $i \in I^*$, so $\tilde{r}_x = \tilde{r}_y$. The same holds for u'_0. By Equation (70), $\tilde{r}_x + \tilde{r}_y = \tilde{r}'_{x'} + \tilde{r}'_{y'}$, hence $\tilde{r}_x = \tilde{r}_y = \tilde{r}'_{x'} = \tilde{r}'_{y'}$. Together with Equation (69), this gives a vertex bijection π satisfying $\tilde{r}'_{\pi(v)} = \tilde{r}_v$ for every $v \in V(G)$. Hence $G \cong G'$ by Equation (57).

Case B2 with $|\operatorname{supp} u_0| = 4$. Both kernel vectors have support of size four and sum zero. Their outside rows agree by Equation (69), and their support-row sums agree by Equation (70). Thus Lemma 15 identifies the remaining reduced rows as multisets. Combining the two correspondences gives a bijection π with $\tilde{r}'_{\pi(x)} = \tilde{r}_x$, which is an isomorphism by Equation (57). □

Proof of Theorem 2. Suppose $G \equiv_{\mathcal{D}_3} G'$. By Lemma 13, $G \sim_{\mathrm{im}} G'$. Hence $G \cong G'$ by Lemma 16. Conversely, an isomorphism $\pi : G \to G'$ gives a bijection $\varphi \mapsto \pi \circ \varphi$ between homomorphisms $F \to G$ and $F \to G'$ for every F, so $G \equiv_{\mathcal{D}_3} G'$. A graph with simple spectrum has nullity at most one, because a real symmetric matrix with pairwise distinct eigenvalues has zero as an eigenvalue at most once; hence $\mathcal{D}_3$ is complete on simple spectrum. □

Remark 17. The nullity bound in Theorem 2 cannot be raised to three. Let B be the cricket graph shown in Figure 6. The graphs $\mathrm{CFI}_0(B)$ and $\mathrm{CFI}_1(B)$ defined in Equation (2) each have 14 vertices. A direct calculation gives nullity three for both graphs. They are non-isomorphic and homomorphism indistinguishable over $\mathcal{D}_3$ by [15, Corollary 3.7 and the proof of Corollary 4.2]. It remains open whether Theorem 2 extends to graphs of nullity two. An exhaustive search found no pair of non-isomorphic graphs of nullity two on at most 11 vertices that are homomorphism indistinguishable over $\mathcal{D}_3$.

4 Proof of Proposition 3

It suffices to prove the proposition for a subsequence of (H_N). If $\mathcal{F}' \subseteq \mathcal{F}$ is the class obtained from a subsequence and $\equiv_{\mathcal{F}'}$ coincides with $\equiv_{\mathcal{D}_3}$, then $\equiv_{\mathcal{F}}$ coincides with $\equiv_{\mathcal{D}_3}$ as well, because $\mathcal{F}' \subseteq \mathcal{F} \subseteq \mathcal{D}_3$. Every tail of (H_N) still has unbounded clique-minor size, since a finite initial segment has bounded clique-minor size. We can therefore choose and reindex a subsequence so that H_N contains K_{3N} as a minor for every $N \geq 1$; Lemma 18 below uses this property. We use the subdivision notation $H^{\mathbf{m}}$ and the labelled coefficients $c_G(H, \ell)$ from Section 3. For this reindexed sequence, let $\mathcal{F}'$ consist of K_1 and the graphs $H_N^{\mathbf{m}}$ for all $N \geq 1$ and all tuples $\mathbf{m}$.

Lemma 18. *Let $N \geq 1$ and suppose that H_N contains K_{3N} as a minor. Let H be a connected cubic multigraph with at most N vertices. Then H_N contains a subgraph isomorphic to a subdivision of H.*

Proof. Since a subdivision of a subdivision of H is a subdivision of H, it suffices to find a simple subcubic subdivision J of H with at most $3N$ vertices. Such a J is a subgraph of K_{3N} and hence a minor of H_N. Since J is subcubic, it is a topological minor of H_N [6, Section 1.7], so H_N contains a subgraph isomorphic to a subdivision of J and hence to a subdivision of H. Let $J = H^{\mathbf{m}}$, where $m_e = 3$ for each loop, $m_e = 2$ for all but one edge in each group of parallel non-loop edges, and $m_e = 1$ for all other edges. Then J is simple and subcubic. There are two new vertices for each loop and $k - 1 \leq 2$ for each group of $k \geq 2$ parallel non-loop edges. Since H is cubic, these loops and parallel-edge groups have disjoint vertex sets. Hence $|V(J)| - |V(H)| \leq 2|V(H)| \leq 2N$, so J has at most $3N$ vertices. □

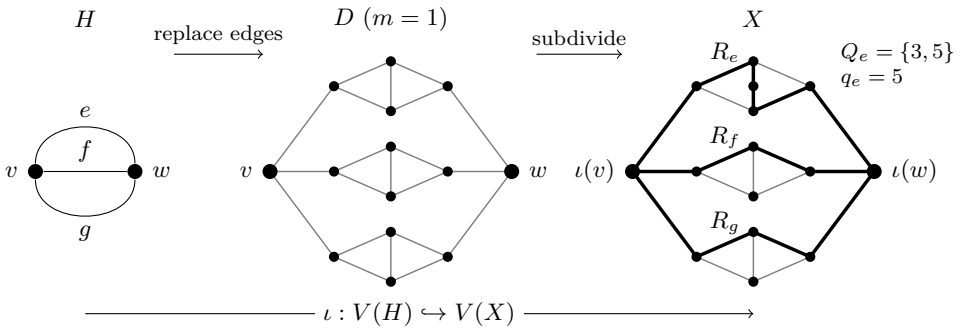


Figure 7: The construction in Lemma 19. Each edge of H is replaced by a diamond in D. In X, the vertical edge of the top diamond is subdivided once. The bold paths form the chosen subdivision of H. The paths for e have four or six edges, so Q_e lists their possible internal-vertex counts.

Lemma 19. *Let H be a connected cubic multigraph and let $m \geq 1$ be an integer. There is a connected cubic graph D with the following property. If a graph X contains a subgraph isomorphic to a subdivision of D, then there are an injection $\iota : V(H) \hookrightarrow V(X)$ and finite sets $Q_e \subseteq \mathbb{Z}_{\geq 1}$ with $|Q_e| \geq m + 1$ for each $e \in E(H)$ such that, for every $\mathbf{q} \in \prod_e Q_e$, there is a subdivision of H in X with vertex map ι and paths R_e of lengths $q_e + 1$. Here R_e is a cycle through $\iota(v)$ if e is a loop at v. At least one edge of X lies outside the chosen subdivision.*

Proof. We construct D by replacing each edge of H by a chain of m diamonds. The *diamond* $K_4 - ab$ is K_4 on vertices a, b, c, d with the edge ab deleted. Its vertices a, b have degree two, and c, d have degree three. A chain consists of m disjoint copies together with an edge from b in each copy to a in the next. For each edge vw of H, take a separate chain and add edges from v to a in its first copy and from b in its last copy to w. The resulting graph D is connected and cubic. Figure 7 illustrates an example of the construction.

Every subdivision of a diamond has two a–b paths of different lengths. To see this, let $\mathrm{len}(acdb)$ denote the length of the path through a, c, d, b, and use the same notation for the other paths. Then

$$\mathrm{len}(acdb) + \mathrm{len}(adcb) = \mathrm{len}(acb) + \mathrm{len}(adb) + 2\,\mathrm{len}(cd). \tag{72}$$

Since $\mathrm{len}(cd) > 0$, these four paths cannot all have the same length.

Fix an isomorphism from a subdivision of D onto a subgraph of X, and let ι be its restriction to $V(H)$. In each subdivided diamond in its image, choose two a–b paths of different lengths. For an edge $e = vw$ of H, concatenate one chosen path from each diamond in its chain with the subdivided joining edges, including those incident with $\iota(v)$ and $\iota(w)$, to obtain R_e. Switching from the shorter to the longer path in one copy at a time gives at least $m + 1$ distinct lengths for R_e. Let Q_e be the set of values $|E(R_e)| - 1$ over these choices. Every R_e has an internal vertex, so $Q_e \subseteq \mathbb{Z}_{\geq 1}$. The paths for different edges have disjoint interiors, all avoiding $\iota(V(H))$. Thus every $\mathbf{q} \in \prod_e Q_e$ can be realised by a subdivision of H in X with the same vertex map ι. Every a–b path in a diamond omits an edge of the diamond. The corresponding subdivided edge in X lies outside all chosen paths R_e, proving the last assertion. □

For a connected graph K with at least two vertices, let $\rho_K > 0$ be its spectral radius and let u_K be its positive unit ρ_K-eigenvector. The Perron–Frobenius theorem gives $P_{\rho_K,K}(x, y) = u_K(x)u_K(y)$. Let $\tau_K = \sum_x u_K(x)^3$ and, for every nonzero real λ, let

$$M_{\lambda,K} = P_{\lambda,K}\,\mathrm{diag}(u_K)P_{\lambda,K}. \tag{73}$$

Since u_K has positive entries, $\tau_K > 0$ and $M_{\lambda,K}$ is positive semidefinite. For $0 \neq \xi \in E_{\lambda,K}$, we have $\xi^\top M_{\lambda,K}\xi = \xi^\top \mathrm{diag}(u_K)\xi > 0$. Since $M_{\lambda,K}$ is zero on $E_{\lambda,K}^\perp$, its image is exactly $E_{\lambda,K}$.

The next lemma computes the effect of the additional vertices in a subdivision of H inside a cubic graph X. Summing over the images of internal path vertices gives powers of $M_{\lambda,K}$; summing over the images of vertices outside the subdivision gives powers of τ_K.

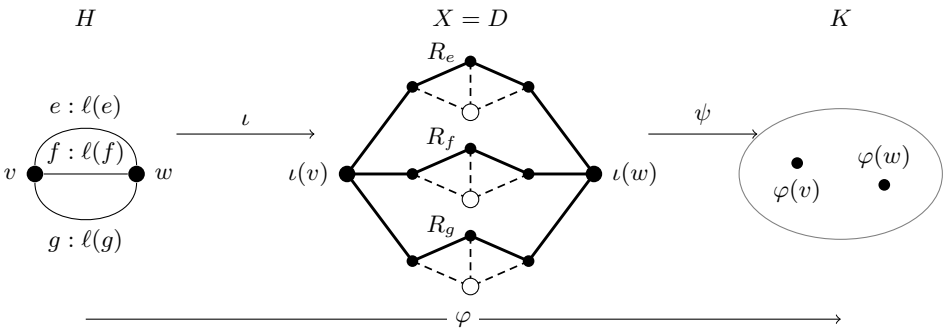


Figure 8: The objects in Lemma 20, using D from Figure 7. Edges of R_e have label $\ell(e)$; each solid path has three internal vertices. Dashed edges have label ρ_K; hollow vertices lie outside all R_e. The target K is schematic, with two vertex images marked. The coefficient $c_K(X, \ell_X)$ sums over all maps $\psi : V(X) \to V(K)$; the right side of Equation (74) sums over all $\varphi : V(H) \to V(K)$. For a fixed ψ, we have $\varphi = \psi \circ \iota$.

Lemma 20. *Let K be a connected graph with at least two vertices, and use the notation of Equation* (73)*. Let H be a cubic multigraph and let X be a cubic graph containing a subdivision of H with vertex map $\iota : V(H) \hookrightarrow V(X)$ and paths $(R_e)_{e \in E(H)}$. For a loop e at v, the path R_e is a cycle through $\iota(v)$. For each edge e of H, let $q_e = |E(R_e)| - 1$ and assume $q_e \geq 1$. For a labelling $\ell : E(H) \to \mathbb{R}^\times$, let ℓ_X assign $\ell(e)$ to the edges of R_e and ρ_K to every other edge of X. Then*

$$c_K(X, \ell_X) = \tau_K^{|V(X)|-|V(H)|} \sum_{\varphi : V(H) \to V(K)} \prod_{e=vw \in E(H)} \left(\frac{M_{\ell(e),K}}{\tau_K} \right)^{q_e} (\varphi(v), \varphi(w)). \tag{74}$$

Figure 8 identifies the graphs, paths, labels and vertex maps in Lemma 20.

Proof. The vertices of X consist of $\iota(V(H))$, the internal vertices of the paths R_e, and the vertices outside all R_e. Since H and X are cubic, the paths occupy all three edge incidences at every vertex of $\iota(V(H))$. At an internal vertex of R_e, two incidences belong to R_e; the third belongs to no chosen path, since their interiors are disjoint. Every edge incident with a vertex outside all R_e lies outside the chosen subdivision.

In the sum defining $c_K(X, \ell_X)$, let $\psi : V(X) \to V(K)$ denote the vertex map and put $\varphi = \psi \circ \iota$. For each edge vw outside all R_e, factor its projector entry as $P_{\rho_K,K}(\psi(v), \psi(w)) = u_K(\psi(v))u_K(\psi(w))$. Regrouping these factors by their endpoints gives no u_K factor at vertices of $\iota(V(H))$, one at each internal path vertex, and three at each vertex outside all R_e. On the paths R_e, retain the projector factors $P_{\ell(e),K}$, including when $\ell(e) = \rho_K$. Summing over the image of each vertex outside all R_e gives $\sum_{x \in V(K)} u_K(x)^3 = \tau_K$. There are $|V(X)| - |V(H)| - \sum_e q_e$ such vertices.

For each edge $e = vw$, let $\lambda = \ell(e)$. Summing over the image of one internal path vertex between fixed neighbour images $x, y \in V(K)$ gives

$$\sum_{z \in V(K)} P_{\lambda,K}(x, z) u_K(z) P_{\lambda,K}(z, y) = M_{\lambda,K}(x, y). \tag{75}$$

Let $x_1, \ldots, x_{q_e}$ be the images of the internal vertices of R_e, and put $x_0 = \varphi(v)$ and $x_{q_e+1} = \varphi(w)$. Since $P_{\lambda,K}^2 = P_{\lambda,K}$ and $q_e \geq 1$, summing over these images gives

$$\sum_{x_1, \ldots, x_{q_e} \in V(K)} \left(\prod_{j=0}^{q_e} P_{\lambda,K}(x_j, x_{j+1}) \right) \left(\prod_{j=1}^{q_e} u_K(x_j) \right) = M_{\lambda,K}^{q_e}(\varphi(v), \varphi(w)). \tag{76}$$

For a loop e, we have $\varphi(v) = \varphi(w)$, so the same calculation gives a diagonal entry of $M_{\lambda,K}^{q_e}$. Summing over φ now gives

$$c_K(X, \ell_X) = \tau_K^{|V(X)|-|V(H)|-\sum_e q_e} \sum_{\varphi : V(H) \to V(K)} \prod_{e=vw \in E(H)} M_{\ell(e),K}^{q_e}(\varphi(v), \varphi(w)). \tag{77}$$

Substituting $M_{\ell(e),K}^{q_e} = \tau_K^{q_e}(M_{\ell(e),K}/\tau_K)^{q_e}$ in Equation (77) proves Equation (74). The total exponent of τ_K is $|V(X)| - |V(H)|$, independent of the path lengths. □

Lemma 21. *Let $M_1, \dots, M_p$ be positive semidefinite operators on finite-dimensional Euclidean spaces, and let P_i be the orthogonal projector onto $\operatorname{im} M_i$. Suppose their spectra together contain at most s distinct positive values, and let Q be a set of at least s distinct positive integers. Then there are coefficients $(a_q)_{q\in Q}$, only finitely many nonzero, with $\sum_{q\in Q} a_q M_i^q = P_i$ for every i.*

Proof. Let $\mu_1, \dots, \mu_{s'}$ be the distinct positive values in the spectra of $M_1, \dots, M_p$. Since $s' \le s$, we can choose distinct elements $q_1, \dots, q_{s'}$ of Q. The generalised Vandermonde matrix $(\mu_h^{q_j})_{h,j=1}^{s'}$ is invertible [8, p. 76]. Hence there are coefficients $a_{q_1}, \dots, a_{q_{s'}}$ with $\sum_j a_{q_j}\mu_h^{q_j} = 1$ for every h; the remaining a_q are zero. The operator $\sum_{q\in Q} a_q M_i^q$ is the identity on each positive eigenspace and is zero on $\ker M_i$, since every q is positive. It therefore equals P_i. □

The next lemma combines two interpolations: one replaces operator powers by image projectors, and the other removes scalar weights.

Lemma 22. *Let $\mathcal{I}, \mathcal{J}$ be finite sets. For each $i \in \mathcal{I}$ and $j \in \mathcal{J}$, let $M_{i,j}$ be a positive semidefinite operator on a finite-dimensional Euclidean space $V_{i,j}$, and let $P_{i,j}$ be the orthogonal projector onto $\operatorname{im} M_{i,j}$. Let Φ_i be a multilinear form on $\prod_{j\in\mathcal{J}} \operatorname{End}(V_{i,j})$, and let $\tau_i > 0$ and $\varepsilon_i \in \mathbb{R}$. Let (d_N) be an unbounded sequence of nonnegative integers. Suppose that, for every sufficiently large N, there are finite sets $Q_{N,j} \subseteq \mathbb{Z}_{\ge 1}$ with $|Q_{N,j}| \ge \sum_{i\in\mathcal{I}} \dim V_{i,j}$, such that*

$$\sum_{i\in\mathcal{I}} \varepsilon_i \tau_i^{d_N} \Phi_i((M_{i,j}^{q_j})_{j\in\mathcal{J}}) = 0 \quad \text{for every } \mathbf{q} \in \prod_{j\in\mathcal{J}} Q_{N,j}. \tag{78}$$

Then $\sum_{i\in\mathcal{I}} \varepsilon_i \Phi_i((P_{i,j})_{j\in\mathcal{J}}) = 0$.

Proof. For each $j \in \mathcal{J}$, the spectra of the operators $(M_{i,j})_{i\in\mathcal{I}}$ together contain at most $\sum_{i\in\mathcal{I}} \dim V_{i,j}$ distinct positive values. Thus, for each sufficiently large N and each j, Lemma 21 gives coefficients $a_{N,j,q}$ with $\sum_{q\in Q_{N,j}} a_{N,j,q} M_{i,j}^q = P_{i,j}$ for every i. By multilinearity,

$$\sum_i \varepsilon_i \tau_i^{d_N} \Phi_i((P_{i,j})_j) = \sum_{\mathbf{q}\in\prod_j Q_{N,j}} \left(\prod_j a_{N,j,q_j}\right) \sum_i \varepsilon_i \tau_i^{d_N} \Phi_i((M_{i,j}^{q_j})_j) = 0. \tag{79}$$

Since d_N is unbounded, we can choose $|\mathcal{I}|$ sufficiently large indices N_h with distinct positive values of d_{N_h}. Applying Lemma 21 to the operators $t \mapsto \tau_i t$ on $\mathbb{R}$ with these exponents gives coefficients b_h with $\sum_h b_h \tau_i^{d_{N_h}} = 1$ for every i. Summing Equation (79) at these indices with coefficients b_h proves the lemma. □

Lemma 23. *Let $(H_N)_{N\ge 1}$ be a sequence of connected cubic graphs such that H_N contains K_{3N} as a minor for every $N \ge 1$. Let G, G' be finite simple graphs. Suppose $c_G(H_N, \ell) = c_{G'}(H_N, \ell)$ for every $N \ge 1$ and every labelling $\ell : E(H_N) \to \mathbb{R}^\times$. Then $c_G(H, \ell) = c_{G'}(H, \ell)$ for every connected cubic multigraph H and every labelling $\ell : E(H) \to \mathbb{R}^\times$.*

Proof. For every graph G, connected multigraph H with an edge, and nonzero edge labelling ℓ, we have

$$c_G(H, \ell) = \sum_K c_K(H, \ell), \tag{80}$$

where K ranges over the connected components of G. Indeed, the spectral projectors of A_G are block diagonal over these components, so a map $V(H) \to V(G)$ contributes zero unless its image lies in one component. For a component K, a label $\lambda > \rho_K$ on any edge gives $c_K(H, \ell) = 0$, because $P_{\lambda,K} = 0$.

Fix an integer $n \ge 0$. We prove the lemma for all pairs G, G' with $|V(G)| + |V(G')| \le 2n$. The combined vertex bound allows the two graphs to have different numbers of vertices after components are deleted. We argue by induction on the total number of components containing an edge. If both graphs are edgeless, including when either has no vertices, every labelled coefficient

of a connected cubic multigraph is zero. Otherwise $n \geq 1$. Let $r > 0$ be the largest spectral radius among all components of both graphs, and let $\mathcal{K}, \mathcal{K}'$ consist of the components of G, G', respectively, with spectral radius r. Since r is attained, $\mathcal{K} \sqcup \mathcal{K}' \neq \varnothing$. Fix a connected cubic multigraph H and a nonzero edge labelling ℓ. We prove that

$$\sum_{K \in \mathcal{K}} c_K(H, \ell) = \sum_{K \in \mathcal{K}'} c_K(H, \ell). \tag{81}$$

Apply Lemma 19 with $m = 2n$ to obtain D. For every sufficiently large N, Lemma 18 places a subdivision of D in H_N. Hence Lemma 19 supplies an injection $\iota : V(H) \hookrightarrow V(H_N)$ and sets $Q_e \subseteq \mathbb{Z}_{\geq 1}$ with $|Q_e| \geq 2n+1$. For each $\mathbf{q} \in \prod_e Q_e$, choose the paths R_e given by that lemma. Let $\ell_N(\mathbf{q})$ assign $\ell(e)$ to the edges of R_e and r to every edge outside the paths. This is the labelling ℓ_X in Lemma 20 with $X = H_N$ and $\rho_K = r$. At least one edge receives r, so only components in $\mathcal{K}$ contribute to $c_G(H_N, \ell_N(\mathbf{q}))$. Put $n_N = |V(H_N)|$ and $n_H = |V(H)|$. Since $\rho_K = r$ for every $K \in \mathcal{K}$, applying Lemma 20 to each such K and using Equation (80) gives

$$c_G(H_N, \ell_N(\mathbf{q})) = \sum_{K \in \mathcal{K}} \tau_K^{n_N - n_H} \sum_{\varphi : V(H) \to V(K)} \prod_{e = vw \in E(H)} \left(\frac{M_{\ell(e),K}}{\tau_K} \right)^{q_e} (\varphi(v), \varphi(w)). \tag{82}$$

The same labelling satisfies Equation (82) for G' and $\mathcal{K}'$ because every $K \in \mathcal{K}'$ also has $\rho_K = r$. We suppress the dependence of the sets Q_e on N. Interpolation over q_e in Equation (82) will replace $(M_{\ell(e),K}/\tau_K)^{q_e}$ by $P_{\ell(e),K}$. Interpolation over N will then replace $\tau_K^{n_N - n_H}$ by 1.

We apply Lemma 22 with component index set $\mathcal{I} = \mathcal{K} \sqcup \mathcal{K}'$ and edge index set $\mathcal{J} = E(H)$. The signs are $+1$ on $\mathcal{K}$ and -1 on $\mathcal{K}'$. For each component K and edge e, the operator is $M_{\ell(e),K}/\tau_K$ on $\mathbb{R}^{V(K)}$. It is positive semidefinite with image projector $P_{\ell(e),K}$. For each component K, let

$$\Phi_K((T_e)_{e \in E(H)}) = \sum_{\varphi : V(H) \to V(K)} \prod_{e = vw \in E(H)} T_e(\varphi(v), \varphi(w)), \tag{83}$$

where each T_e is a matrix indexed by $V(K)$. This form is multilinear, and $\Phi_K((P_{\ell(e),K})_e) = c_K(H, \ell)$. The spaces $\mathbb{R}^{V(K)}$ have total dimension $\sum_{K \in \mathcal{K} \sqcup \mathcal{K}'} |V(K)| \leq 2n$. Every Q_e has at least $2n+1$ elements, so the dimension bound in Lemma 22 holds. The assumed equality of the two instances of Equation (82) is exactly Equation (78) with $d_N = n_N - n_H$. Since H_N contains a K_{3N} minor, $n_N \geq 3N$. Thus d_N is nonnegative for all sufficiently large N and is unbounded. Hence Lemma 22 proves Equation (81).

Let G^- and G'^- be obtained by deleting the components in $\mathcal{K}$ and $\mathcal{K}'$, respectively. Let $N_* \geq 1$ and let η be a nonzero labelling of H_{N_*}. Since H_{N_*} is connected and cubic, Equation (81) holds with $H = H_{N_*}$ and $\ell = \eta$, so by Equation (80) and the assumed equality on H_{N_*},

$$\begin{aligned} c_{G^-}(H_{N_*}, \eta) &= c_G(H_{N_*}, \eta) - \sum_{K \in \mathcal{K}} c_K(H_{N_*}, \eta) \\ &= c_{G'}(H_{N_*}, \eta) - \sum_{K \in \mathcal{K}'} c_K(H_{N_*}, \eta) = c_{G'^-}(H_{N_*}, \eta). \end{aligned} \tag{84}$$

We have $|V(G^-)| + |V(G'^-)| \leq 2n$. Since $\mathcal{K} \sqcup \mathcal{K}' \neq \varnothing$ and $r > 0$, deleting these components strictly decreases the total number containing an edge. The induction hypothesis therefore gives $c_{G^-}(H, \ell) = c_{G'^-}(H, \ell)$ for every connected cubic H and nonzero labelling ℓ. Adding the equal sums in Equation (81) and using Equation (80) gives $c_G(H, \ell) = c_{G'}(H, \ell)$, proving the lemma. □

Proof of Proposition 3. By the initial reduction, it suffices to prove the equivalence for $\mathcal{F}'$. Suppose $G \equiv_{\mathcal{F}'} G'$. Since $K_1 \in \mathcal{F}'$, we have $|V(G)| = |V(G')|$. Their counts agree on every subdivision $H_N^{\mathbf{m}}$. By Lemma 12, $c_G(H_N, \ell) = c_{G'}(H_N, \ell)$ for every N and every nonzero labelling ℓ. Lemma 23 therefore gives equality of the labelled coefficients on every connected cubic multigraph. By Lemma 13(c), $G \equiv_{\mathcal{D}_3} G'$. The reverse implication follows from $\mathcal{F}' \subseteq \mathcal{D}_3$, and completeness on simple spectrum follows from Theorem 2. □

Declarations

The author used OpenAI GPT-6 Astra and Anthropic Claude Opus 5.5 in preparing this work. The author formulated the problem and initial theorem statements. GPT-6 Astra assisted with mathematical exploration and identifying relevant literature. The proofs were developed through discussions between the author and GPT-6 Astra. GPT-6 Astra and Claude Opus 5.5 both assisted with checking readability and improving the presentation. The author assumes responsibility for all content.

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